



SOICT

HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 5.2: Structured Representations and Reasoning

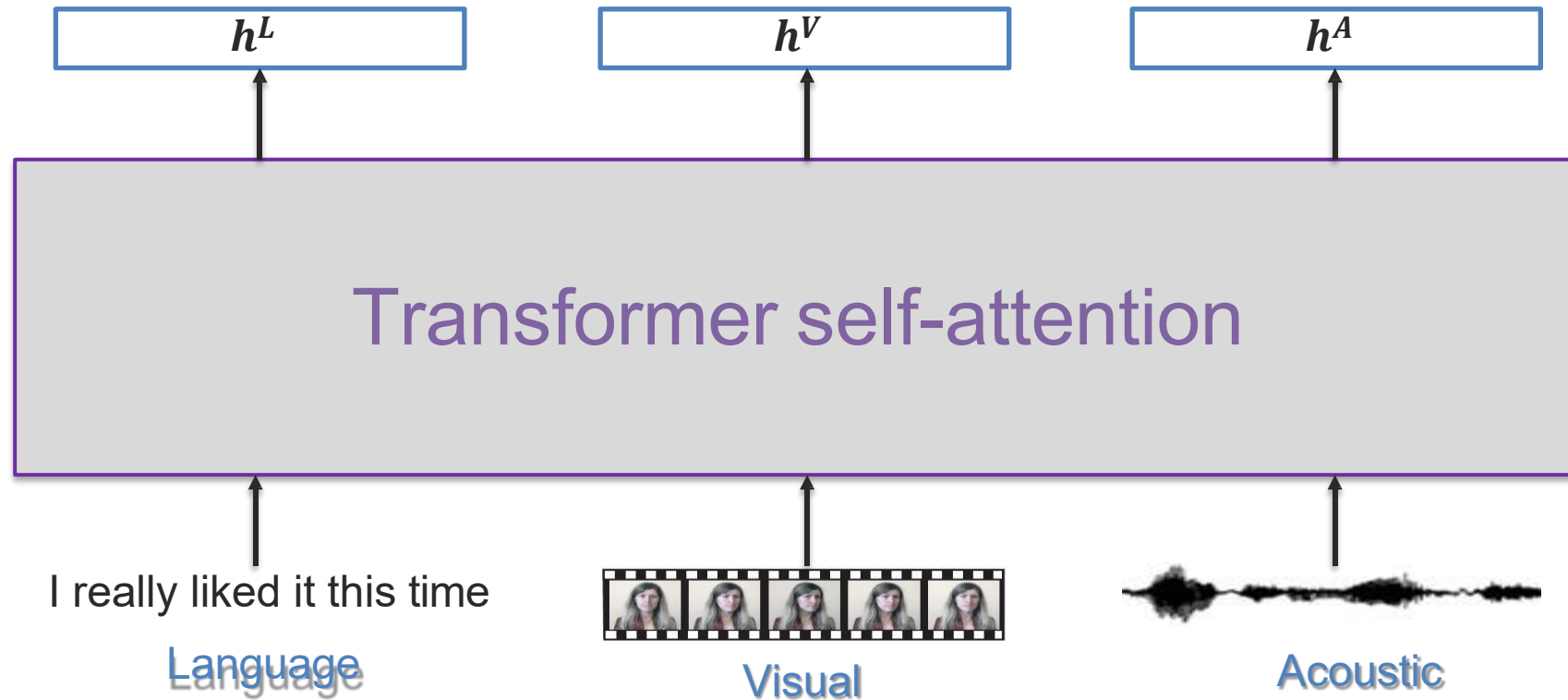
Dam Quang Tuan

Objectives of today's class

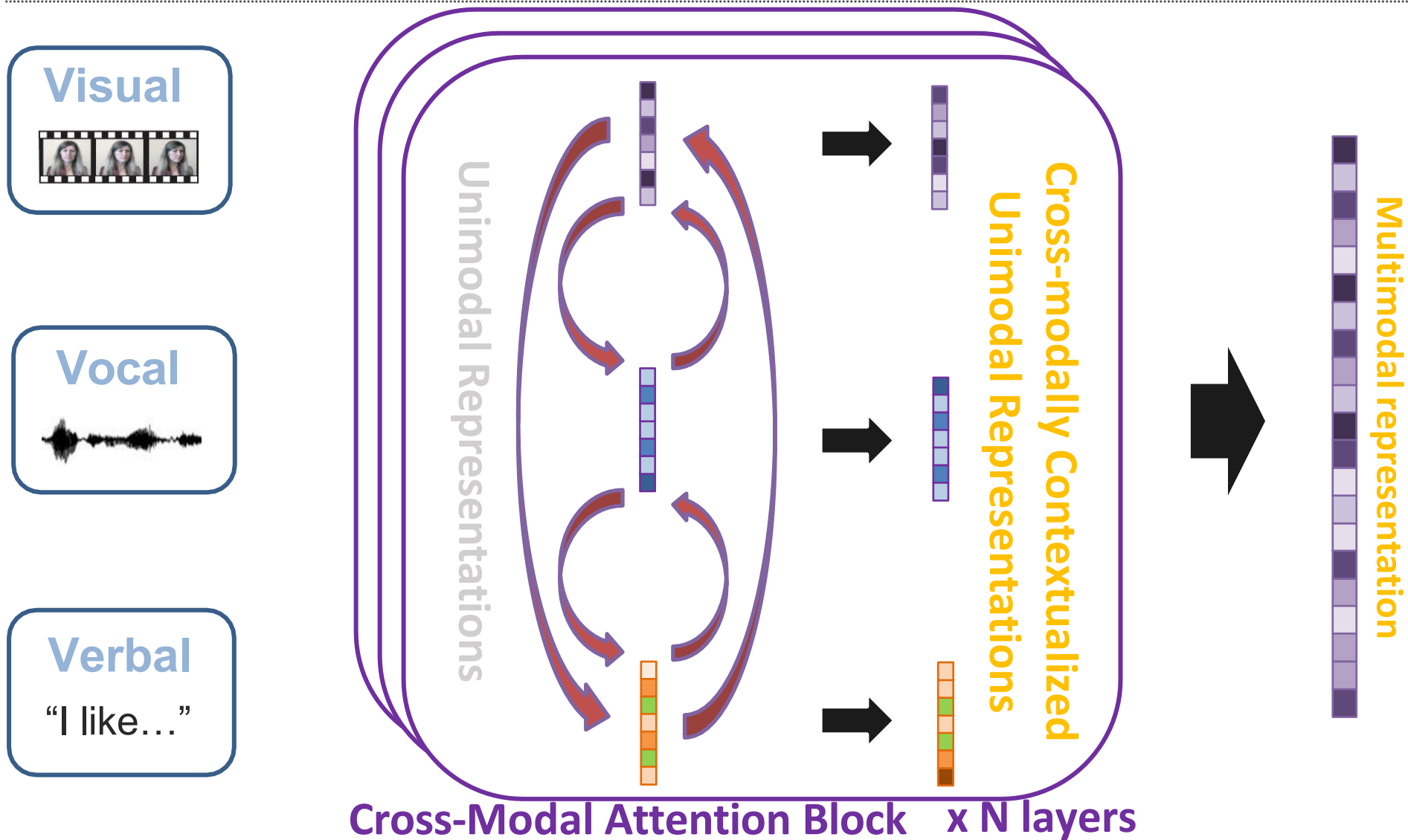
- Multimodal transformers
 - Modality-shift transformer (MAG-BERT)
- Sequence-to-sequence modeling with Transformers
- Going beyond sequences
 - Graph representations
 - Graph neural networks
 - Hierarchical representations
 - Modular representations
 - Neural module networks
 - Neuro-symbolic networks

Language-Vision Transformers

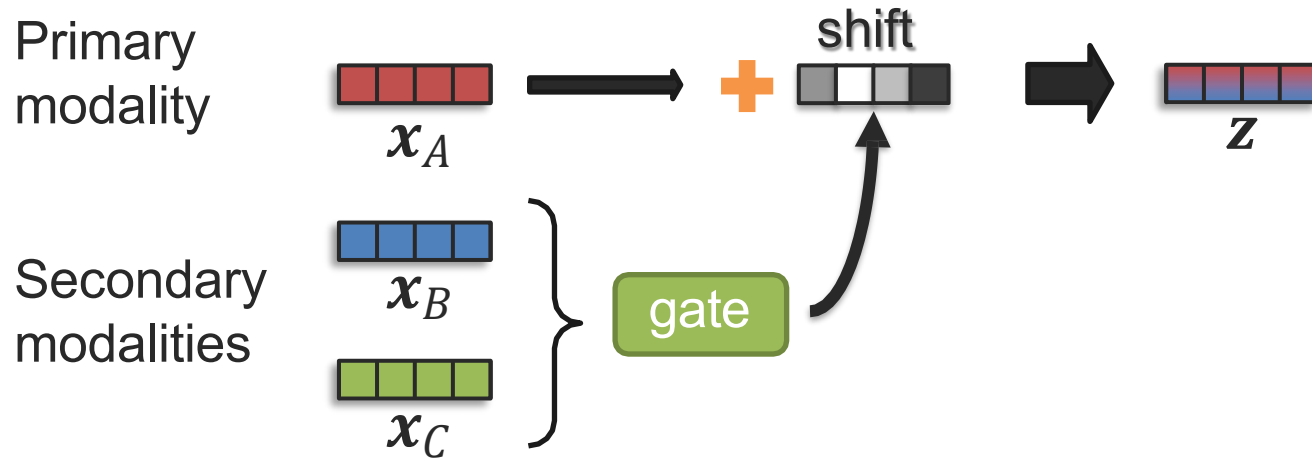
Simple Solution: Contextualized Multimodal Embeddings



Multimodal Transformer – Pairwise Cross-Modal



Reminder: Modality-Shifting Fusion



Example with language modality:

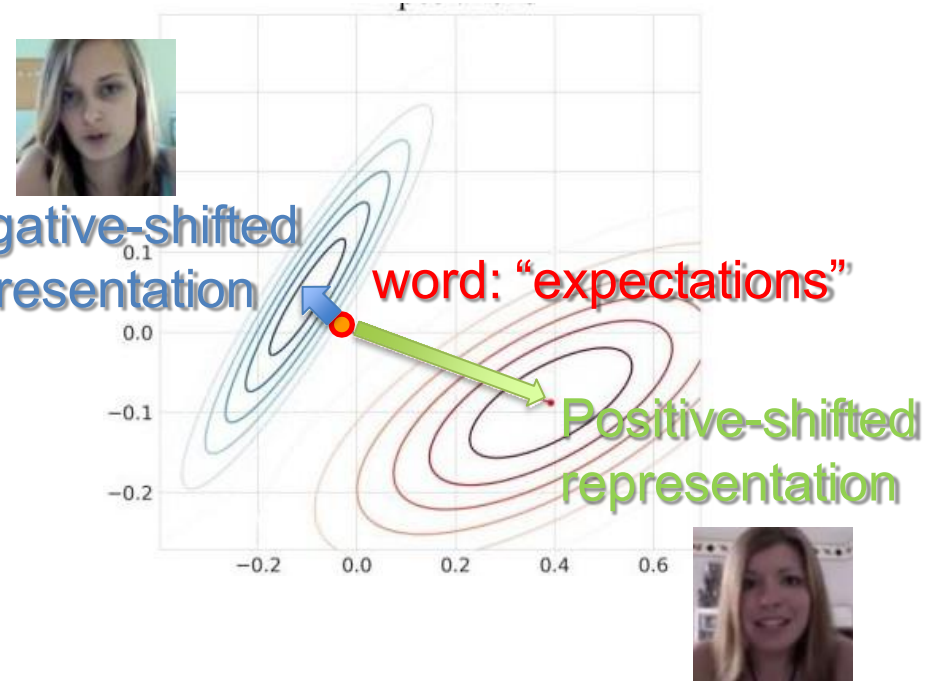
Primary modality: language

Secondary modalities: acoustic and visual

Negative-shifted representation

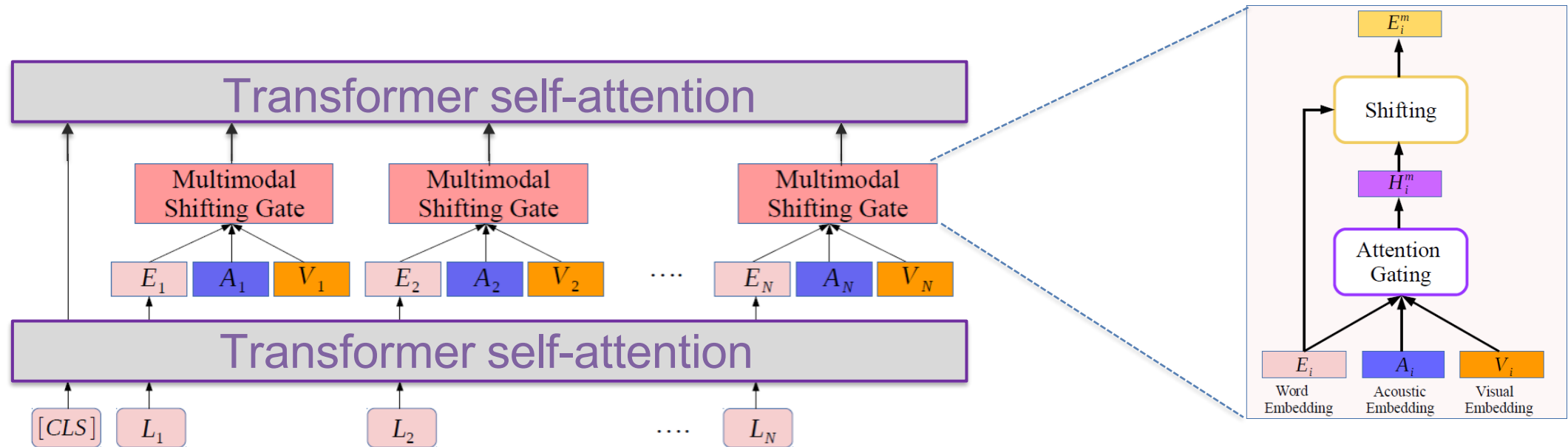
word: "expectations"

Positive-shifted representation



Modality-Shifting with Transformers

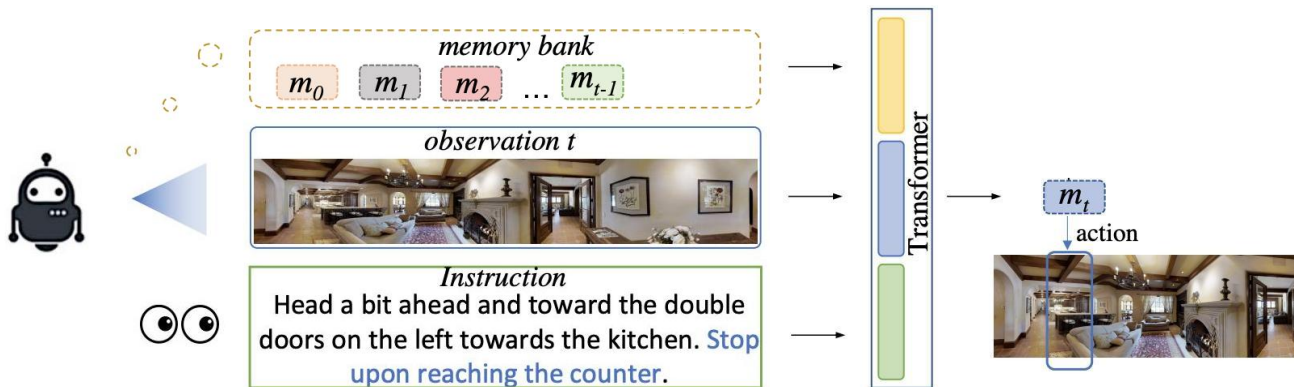
Multimodal Adaptation Gate (MAG) + BERT



Memory for Multimodal Sequences

Memory + aligned contextualized representations

Where have I visited previously?



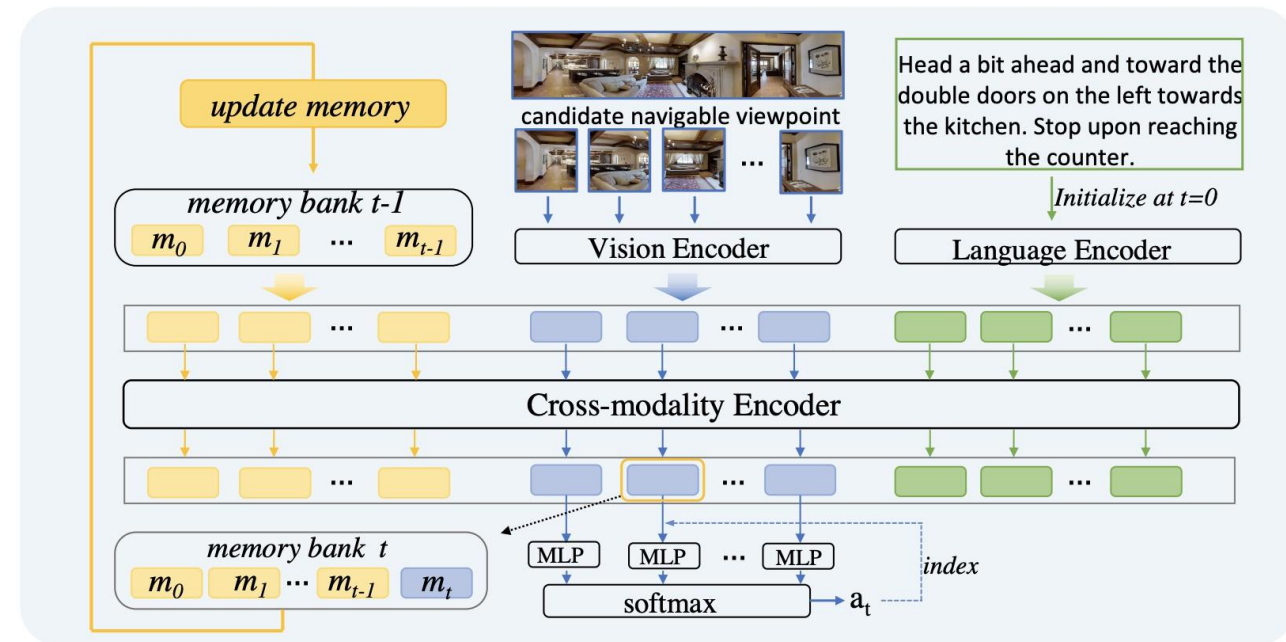
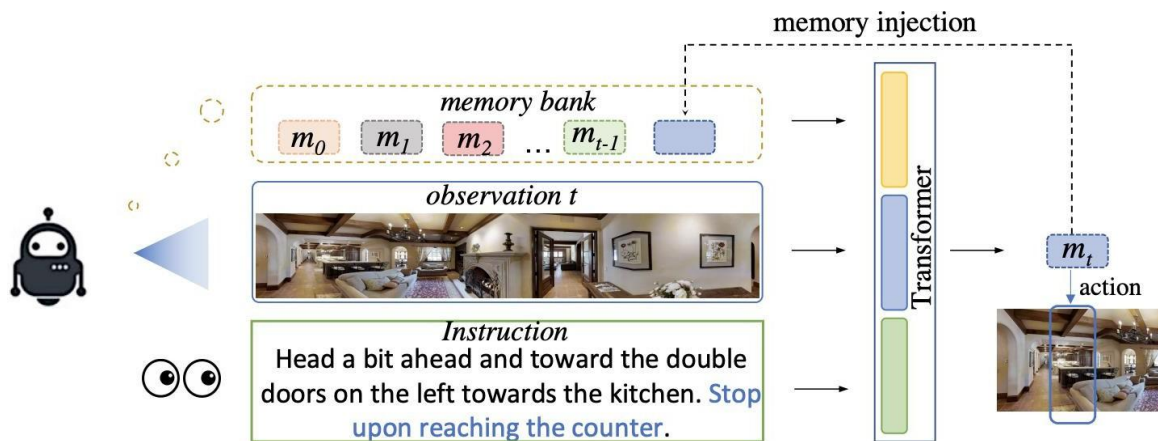
[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]

[Lin et al., Multimodal Transformer with Variable-length Memory for Vision-and-Language Navigation. ECCV 2022]

Memory for Multimodal Sequences

Memory + aligned contextualized representations

Where have I visited previously?



+ Contextualized representations

+ Memory mechanisms

[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]

[Lin et al., Multimodal Transformer with Variable-length Memory for Vision-and-Language Navigation. ECCV 2022]

Memory for Multimodal Sequences

Memory + aligned contextualized representations

Where have I visited previously?

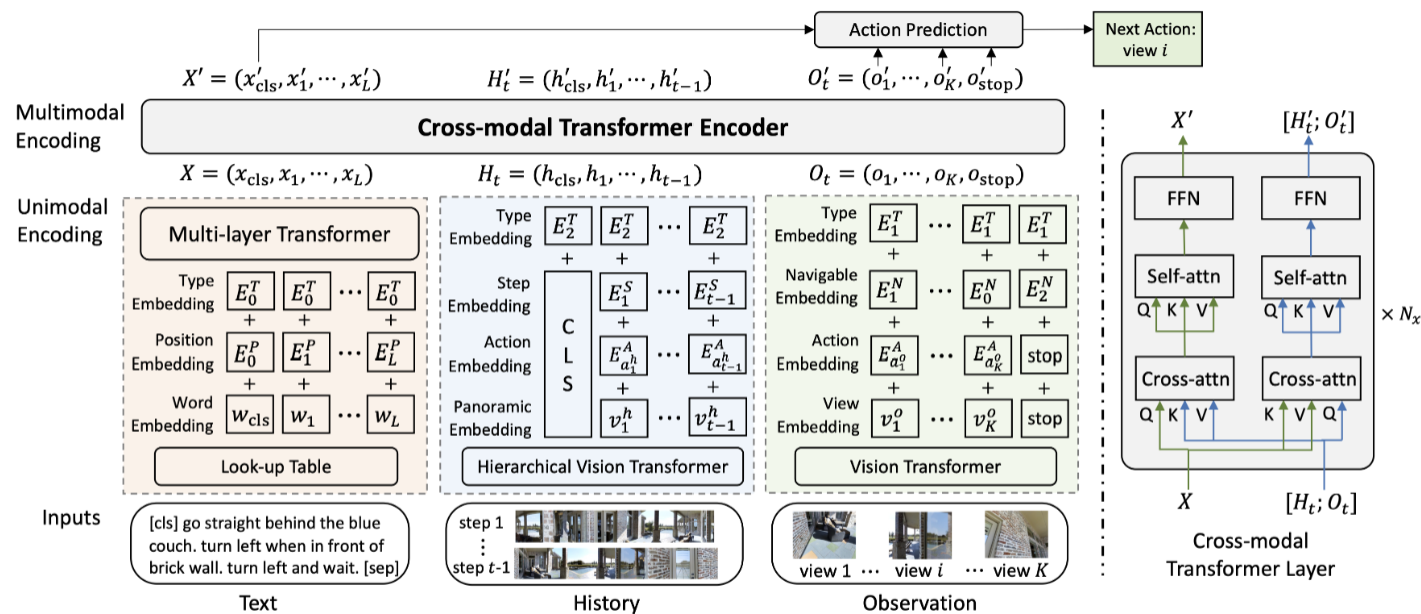
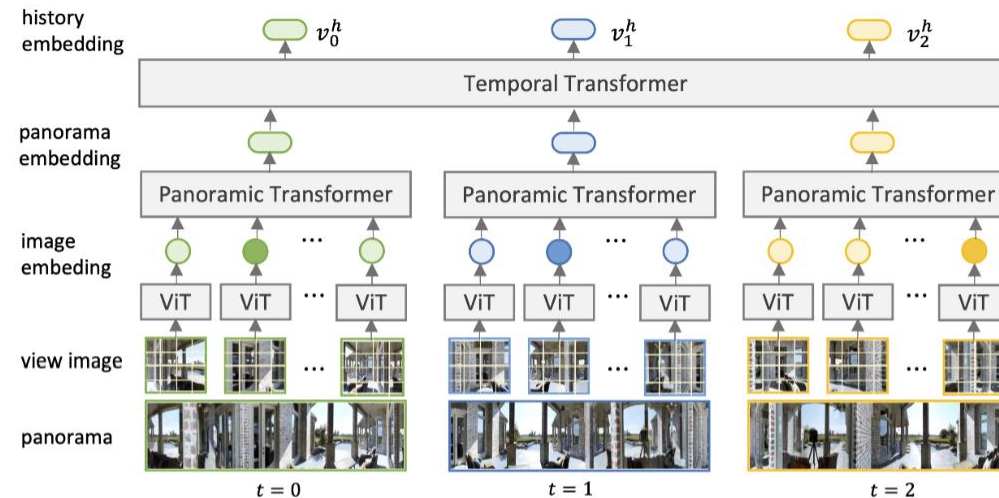


Figure 1: The architecture of History Aware Multimodal Transformer (HAMT). HAMT jointly encodes textual instruction, full history of previous observations and actions, and current observation to predict the next action.

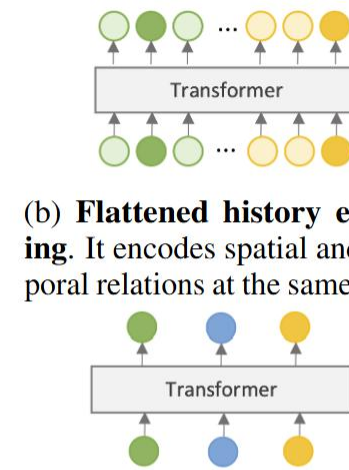
Memory for Multimodal Sequences

Memory + aligned contextualized representations

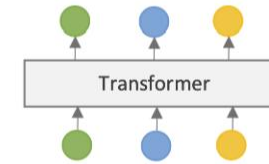
Where have I visited previously?



(a) **Hierarchical history encoding.** It first encodes individual view images with ViT, then models the spatial relation between images in each panorama, and finally captures the temporal relation between panoramas in the history.



(b) **Flattened history encoding.** It encodes spatial and temporal relations at the same time.

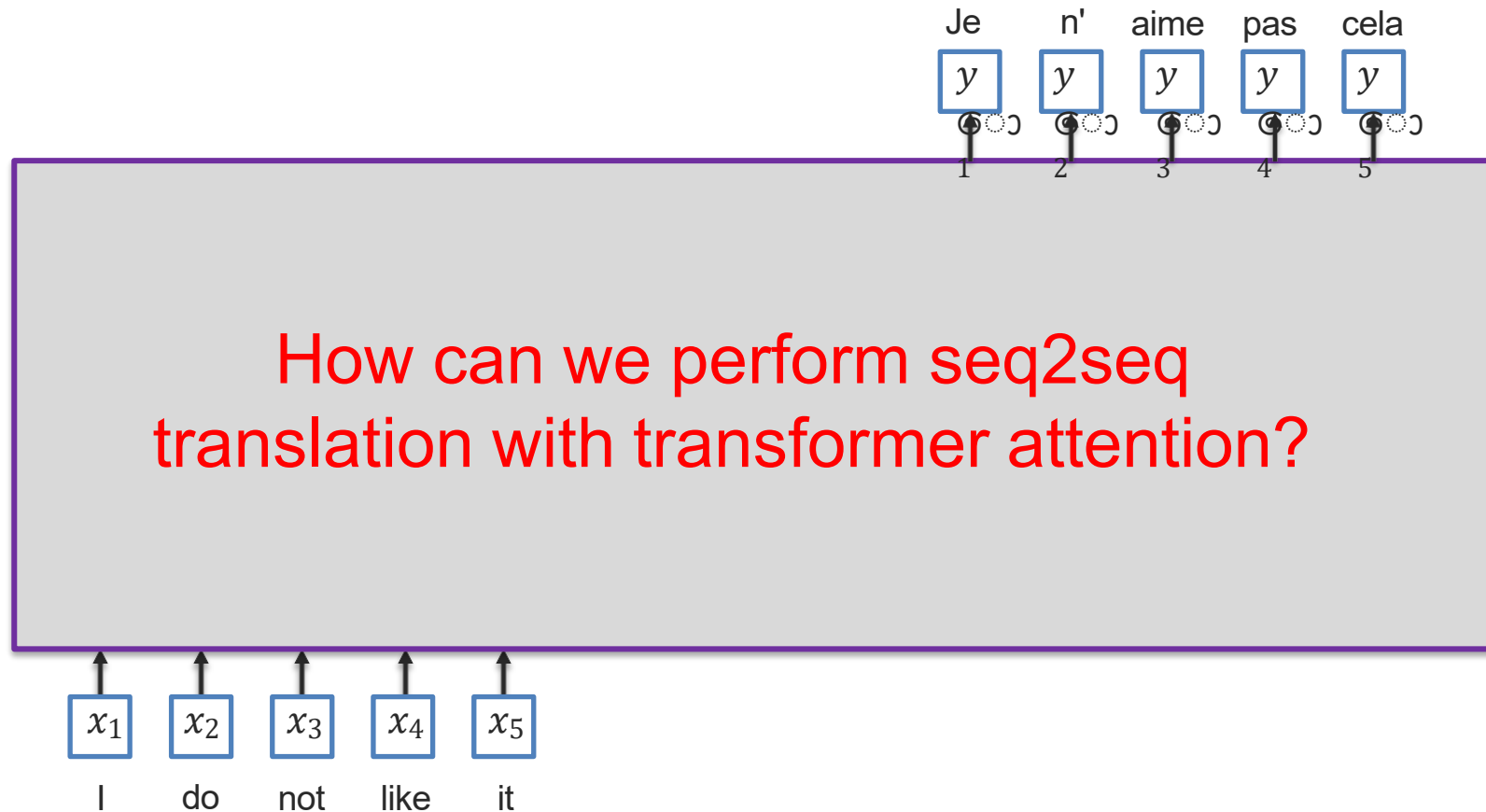


(c) **Temporal-only history encoding.** It only considers temporal relation of oriented views.

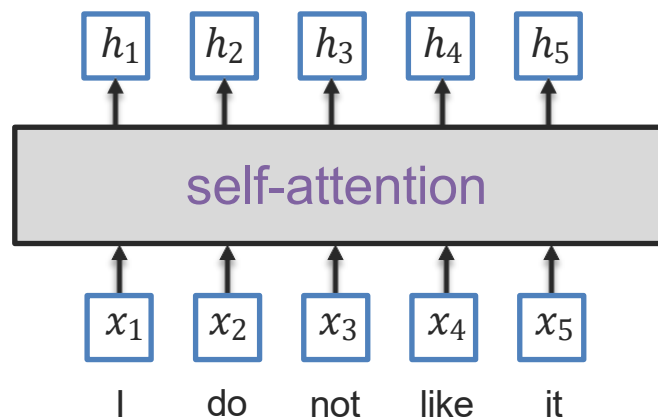
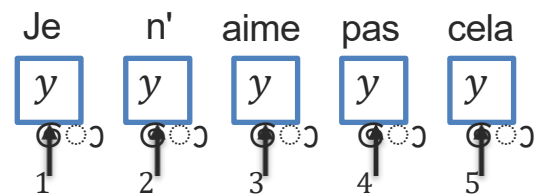
Figure 2: A comparison of history encoding methods. Circle nodes in different colors denote view images of panorama at different steps. Darker circle nodes are the oriented view of the agent.

Sequence-to-Sequence Using Transformer

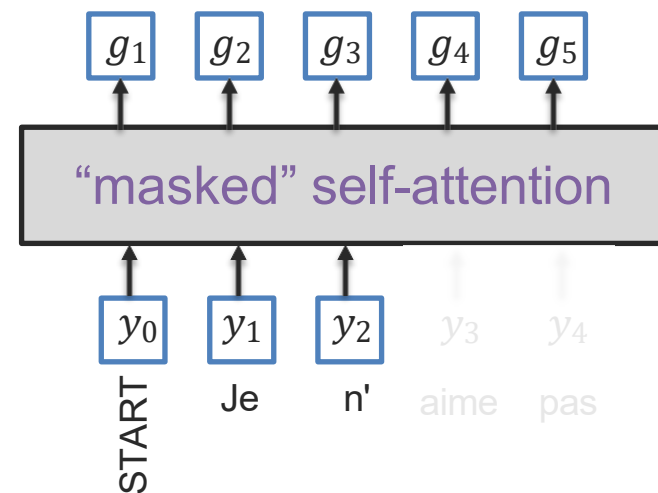
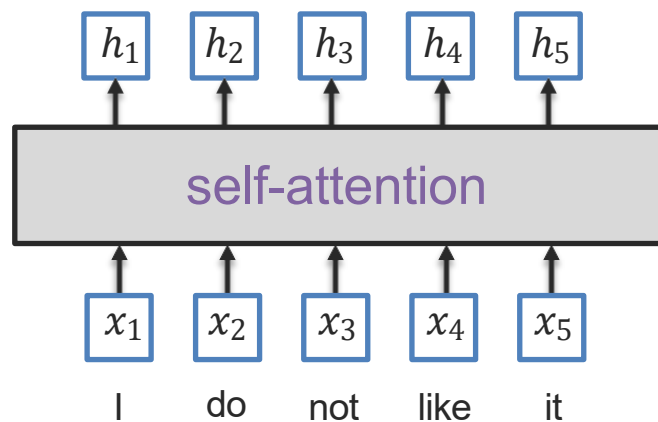
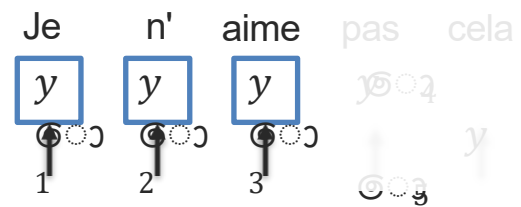
Sequence-to-Sequence Modeling



Seq2Seq with Transformer Attentions

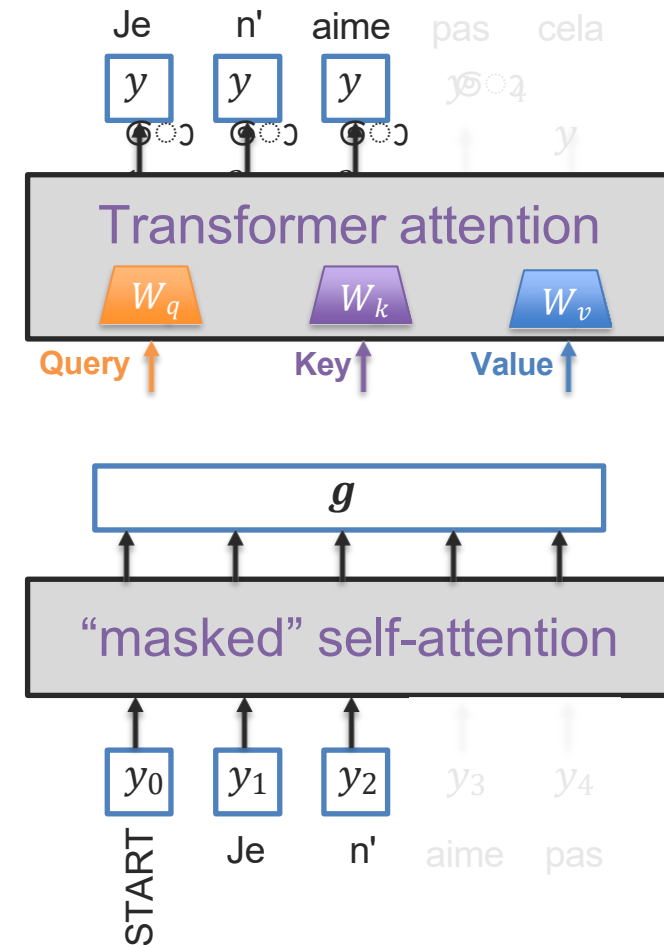
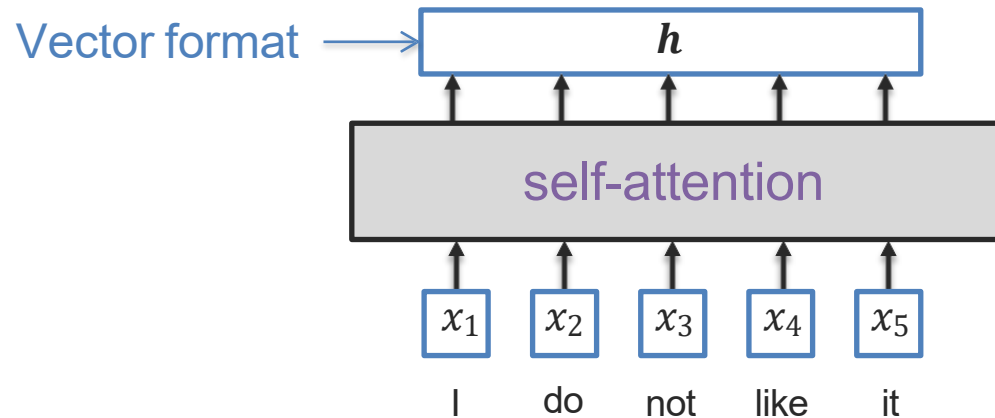


Seq2Seq with Transformer Attentions

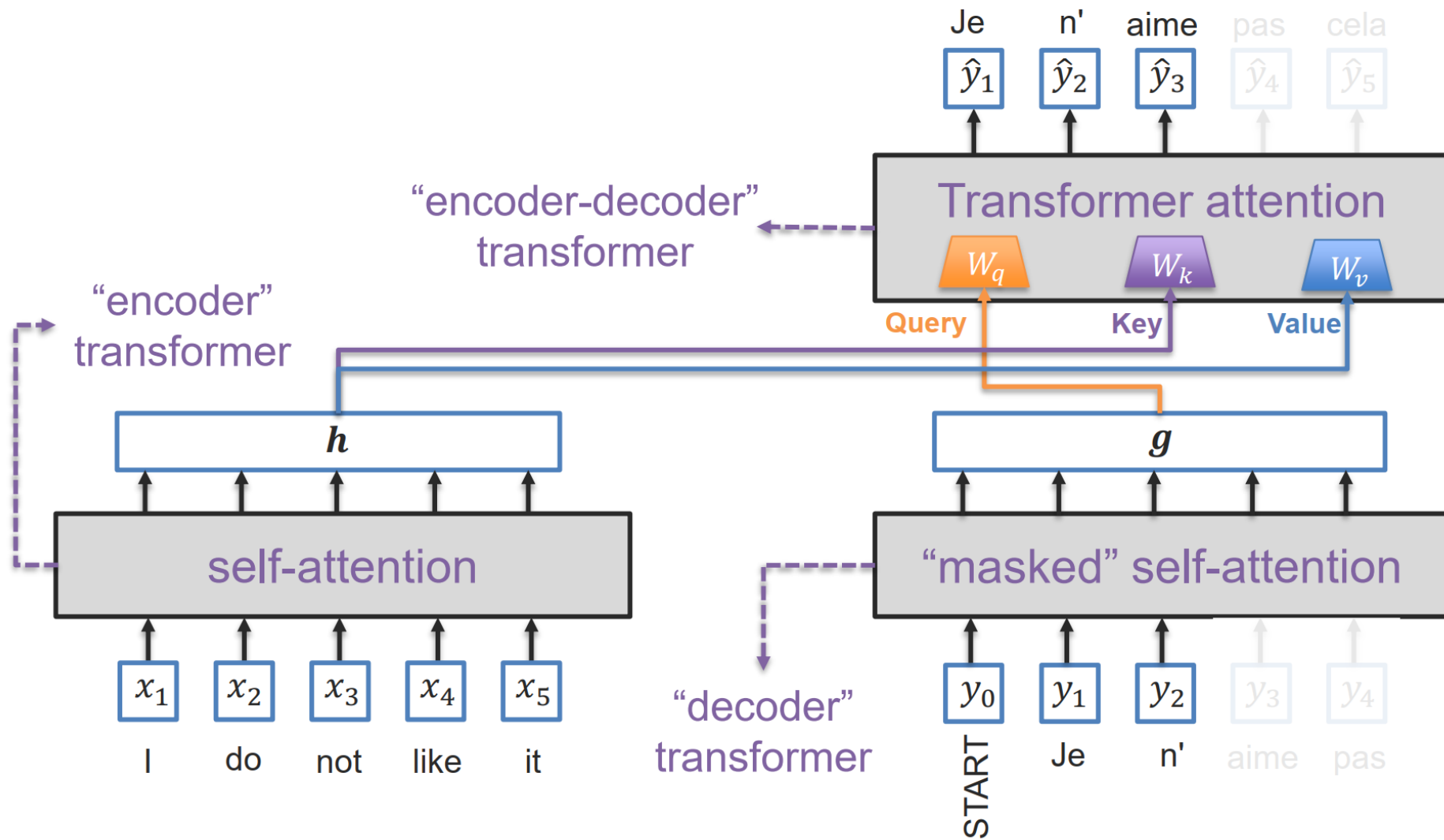


Seq2Seq with Transformer Attentions

How should we connect the encoder and decoder self-attention to the transformer attention?



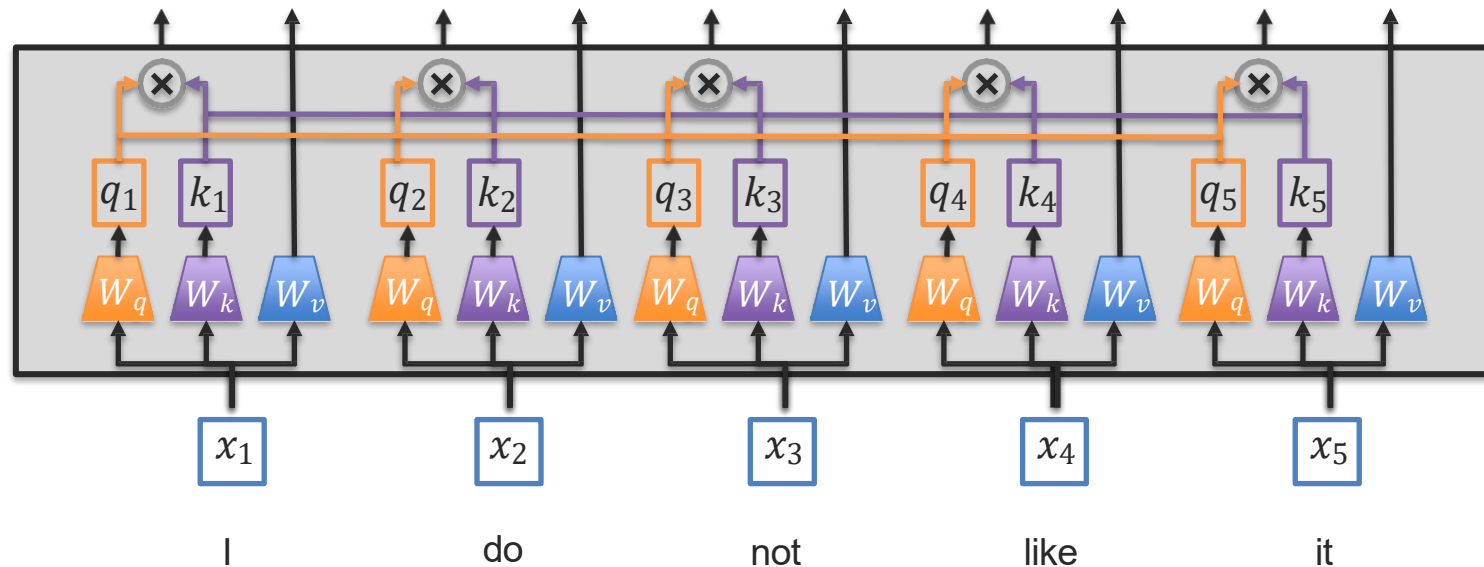
Seq2Seq with Transformer Attentions



Going Beyond Sequences: Graph Representations

*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

Transformers – Fully-Connected Sequences

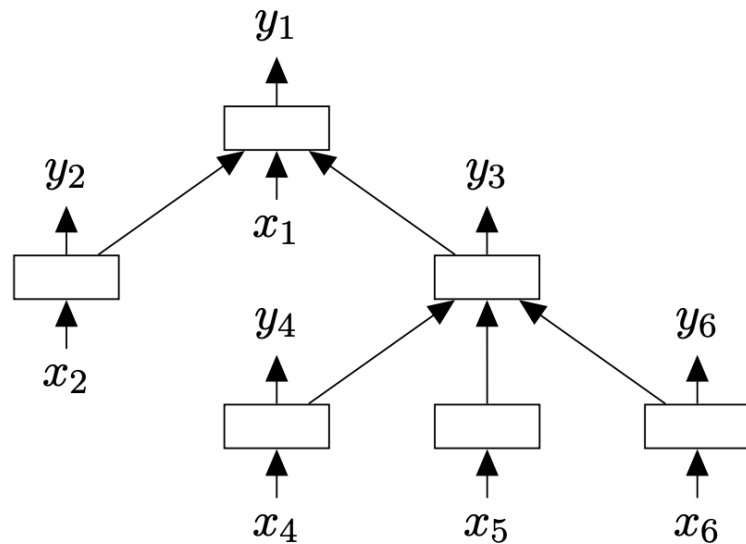
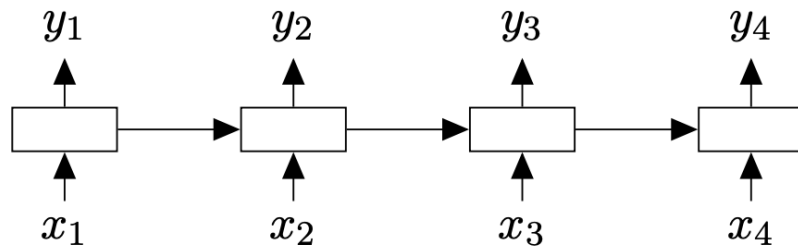


Should everything be connected to everything?

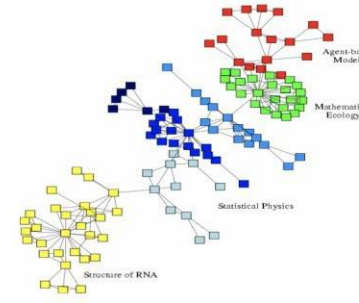
What if we have domain knowledge about connections?

Tree and Graph Networks

From linear chain models to tree and graph-structured models



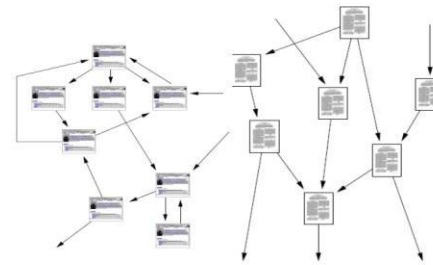
Social networks



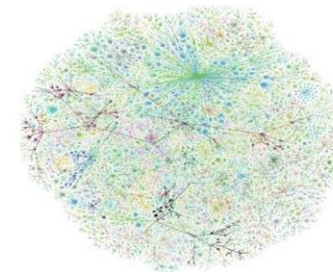
Economic networks



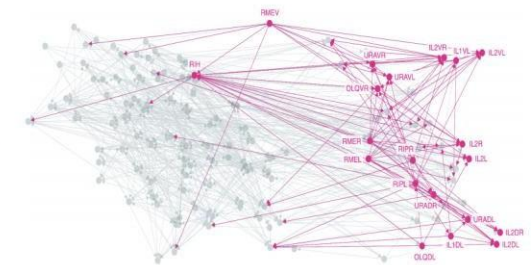
Biomedical networks



Information networks:
Web & citations



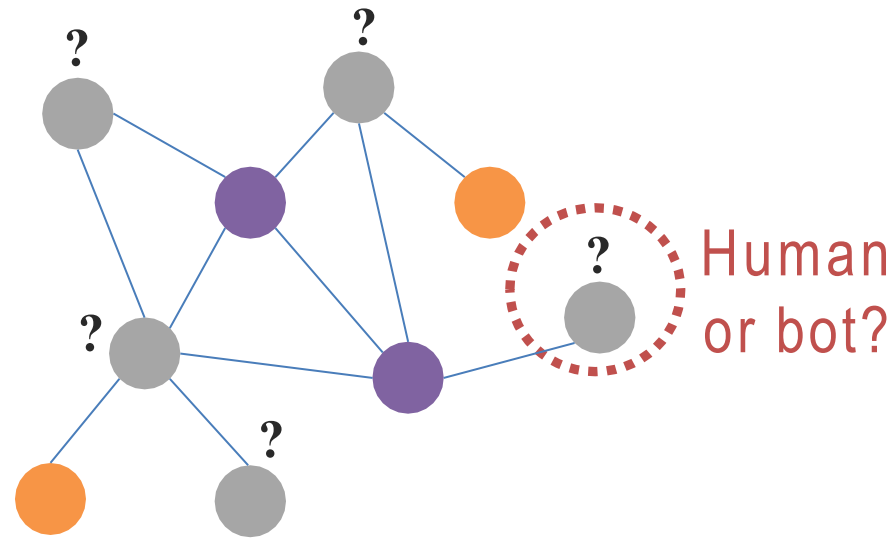
Internet



Networks of neurons

Graphs – Supervised Task

Goal: Learn from labels associated with a subset of nodes (or with all nodes)

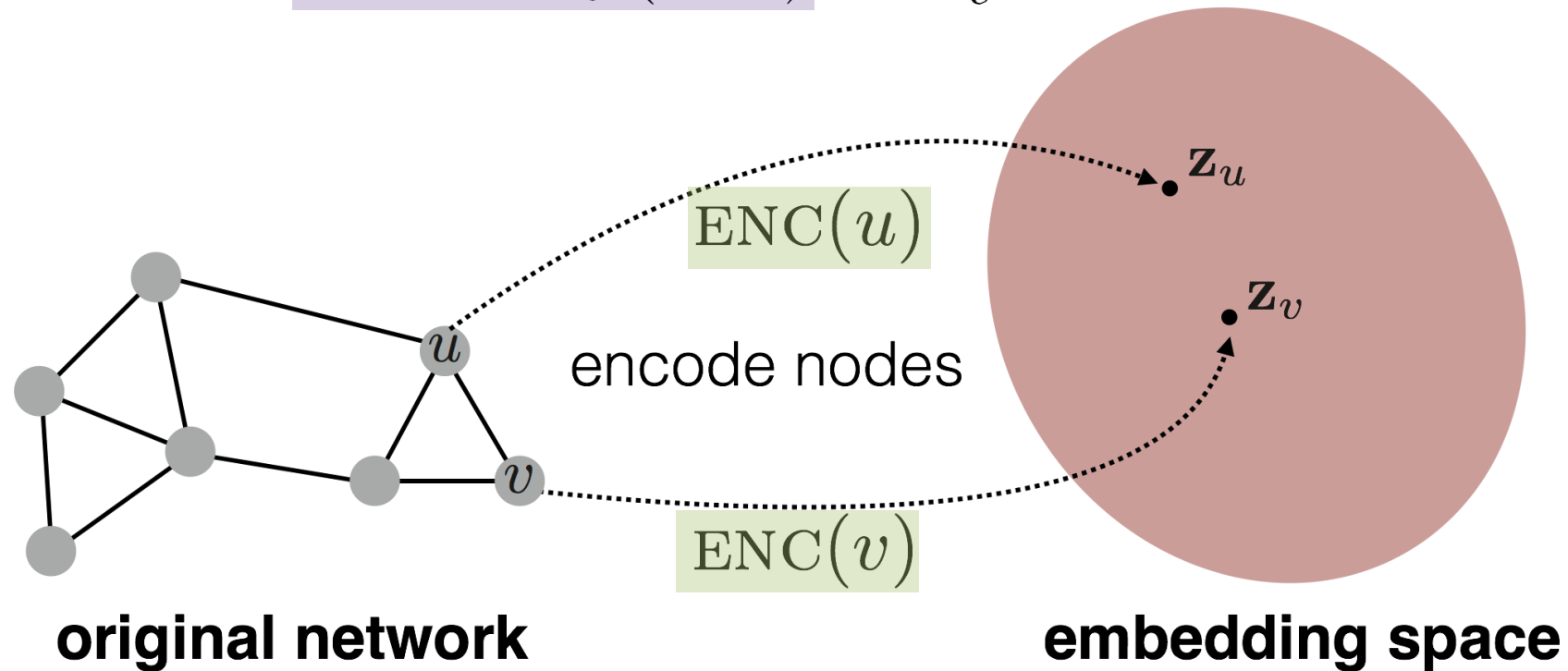


e.g., an online social network

Graphs – Unsupervised Task

Goal: Learn an embedding space where

$$\text{similarity}(u, v) \approx \mathbf{z}_v^\top \mathbf{z}_u$$



Graph Neural Nets

Assume we have a graph $\mathbf{G}$:

$\mathbf{V}$ is the set of vertices

$\mathbf{A}$ is the binary adjacency matrix

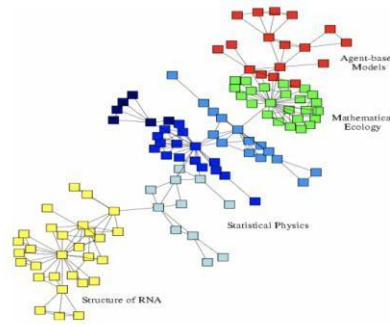
$\mathbf{X}$ is a matrix of node features:

- Categorical attributes, text, image data
e.g. profile information in a social network
- ...

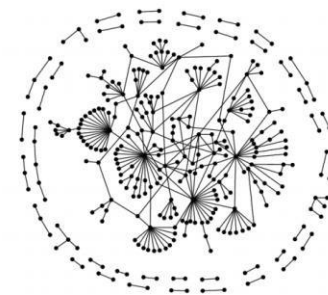
$\mathbf{Y}$ is a vector of node labels (optional)



Social networks



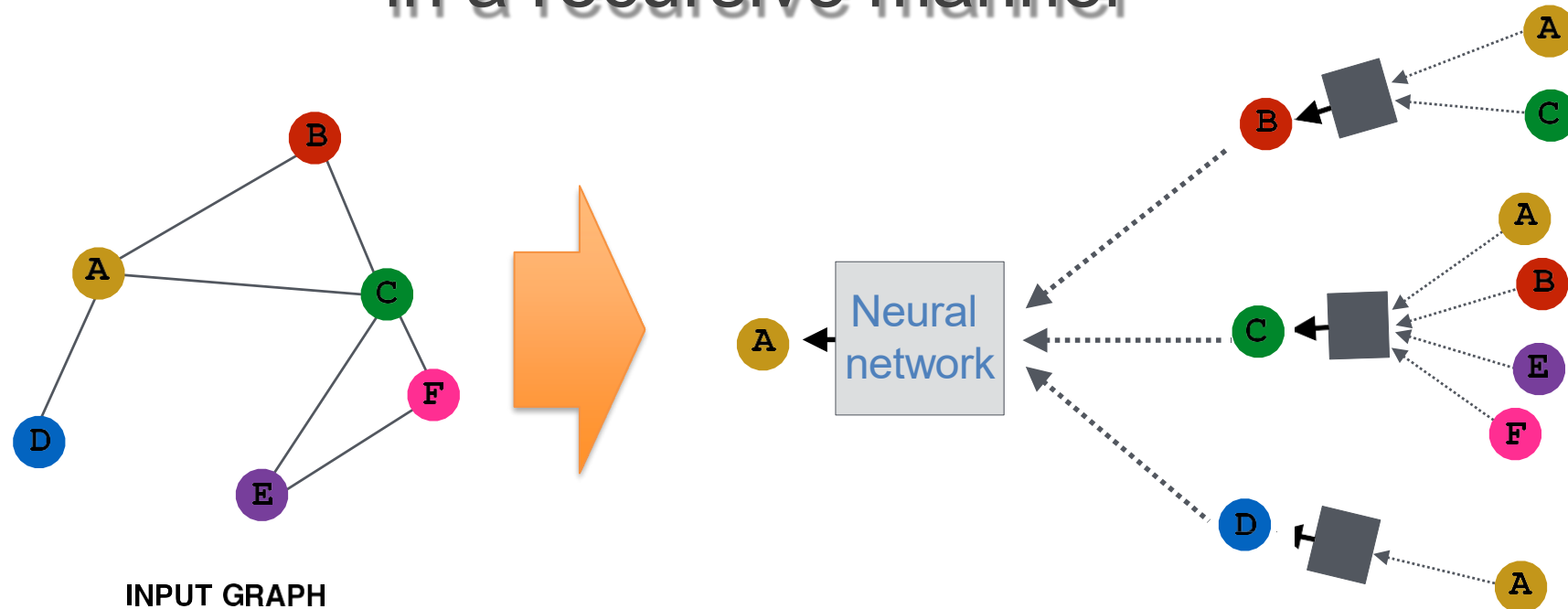
Economic networks



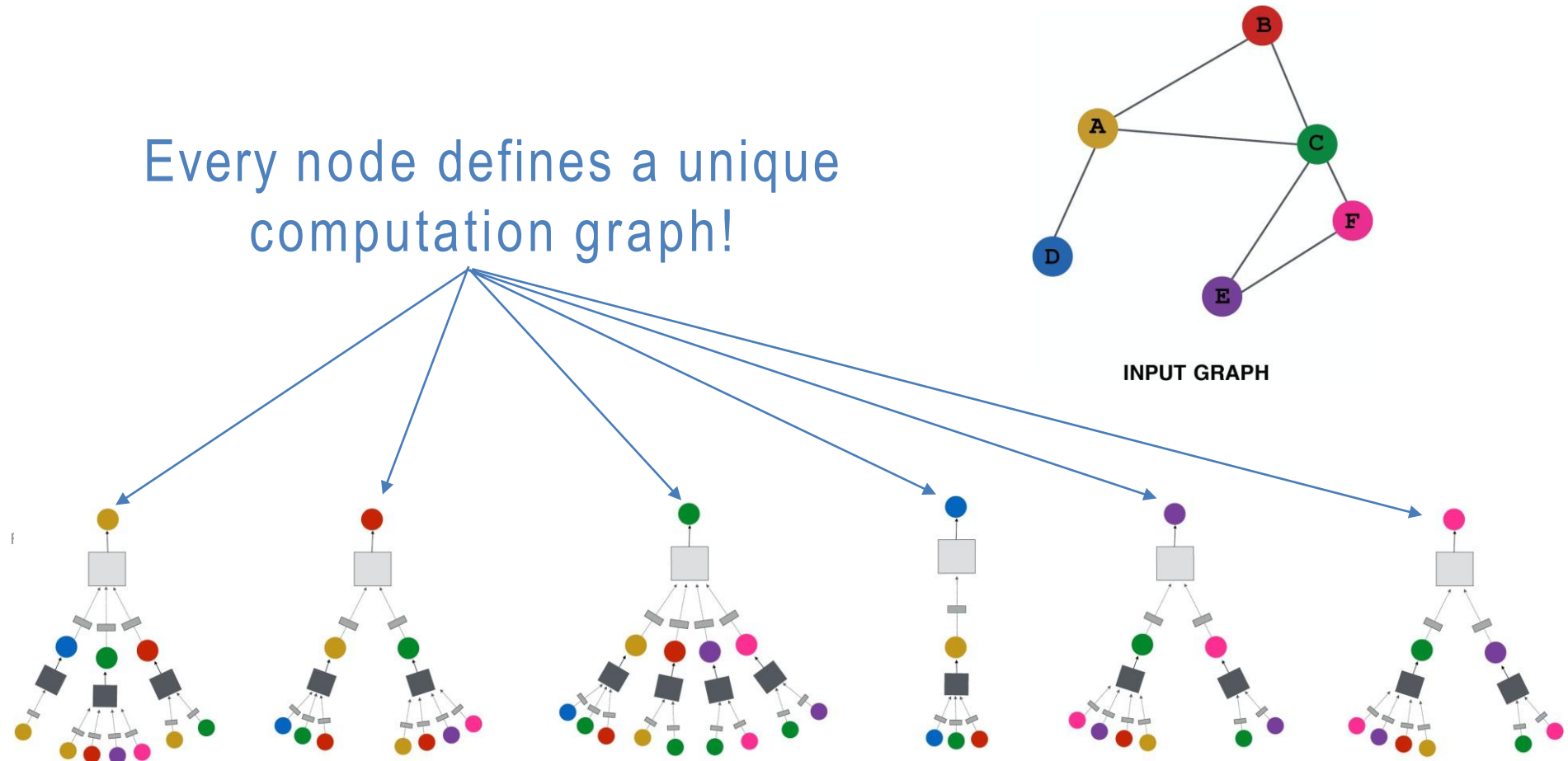
Biomedical networks

Graph Neural Nets

Key idea: Generate node embeddings based on local neighborhoods in a recursive manner



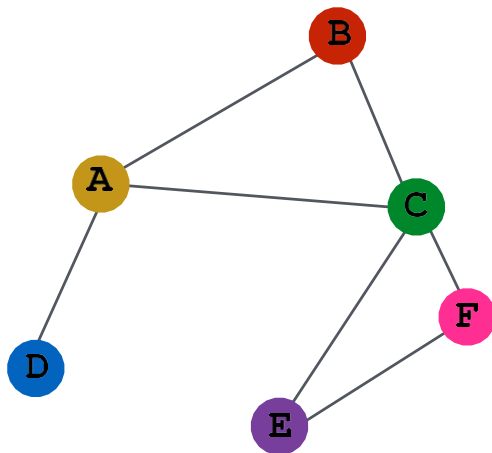
Graph Neural Nets



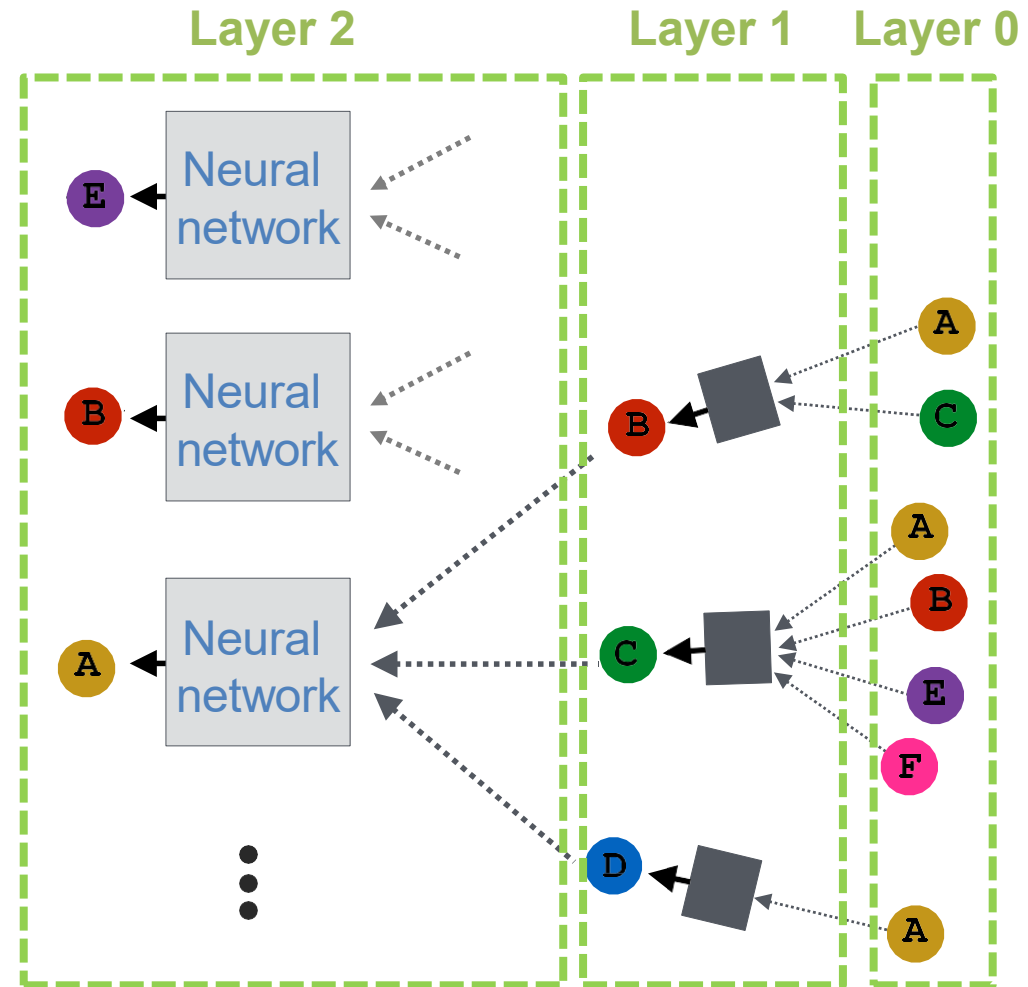
Graph Neural Nets

And multiple layers!

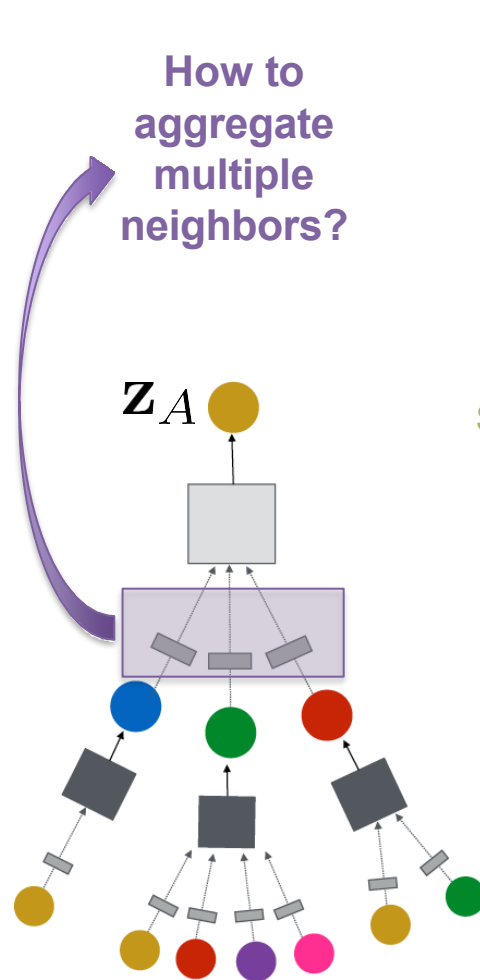
- ➡ Shared parameters within a specific layer
- ➡ “layer-0” is the input feature x_u



INPUT GRAPH



Graph Neural Nets – Neighborhood Aggregation



Average pooling (Scarselli et al., 2005)

$$\mathbf{h}_v^k = \sigma \left(\mathbf{W}_k \sum_{u \in N(v)} \frac{\mathbf{h}_u^{k-1}}{|N(v)|} + \mathbf{B}_k \mathbf{h}_v^{k-1} \right)$$

K is num layers

Different weights for neighbors and self

Graph Convolution Network (Kipf et al., 2017)

$$\mathbf{h}_v^k = \sigma \left(\mathbf{W}_k \sum_{u \in N(v) \cup v} \frac{\mathbf{h}_u^{k-1}}{\sqrt{|N(u)| |N(v)|}} \right)$$

Same weights

Different normalization

It can be efficiently implemented

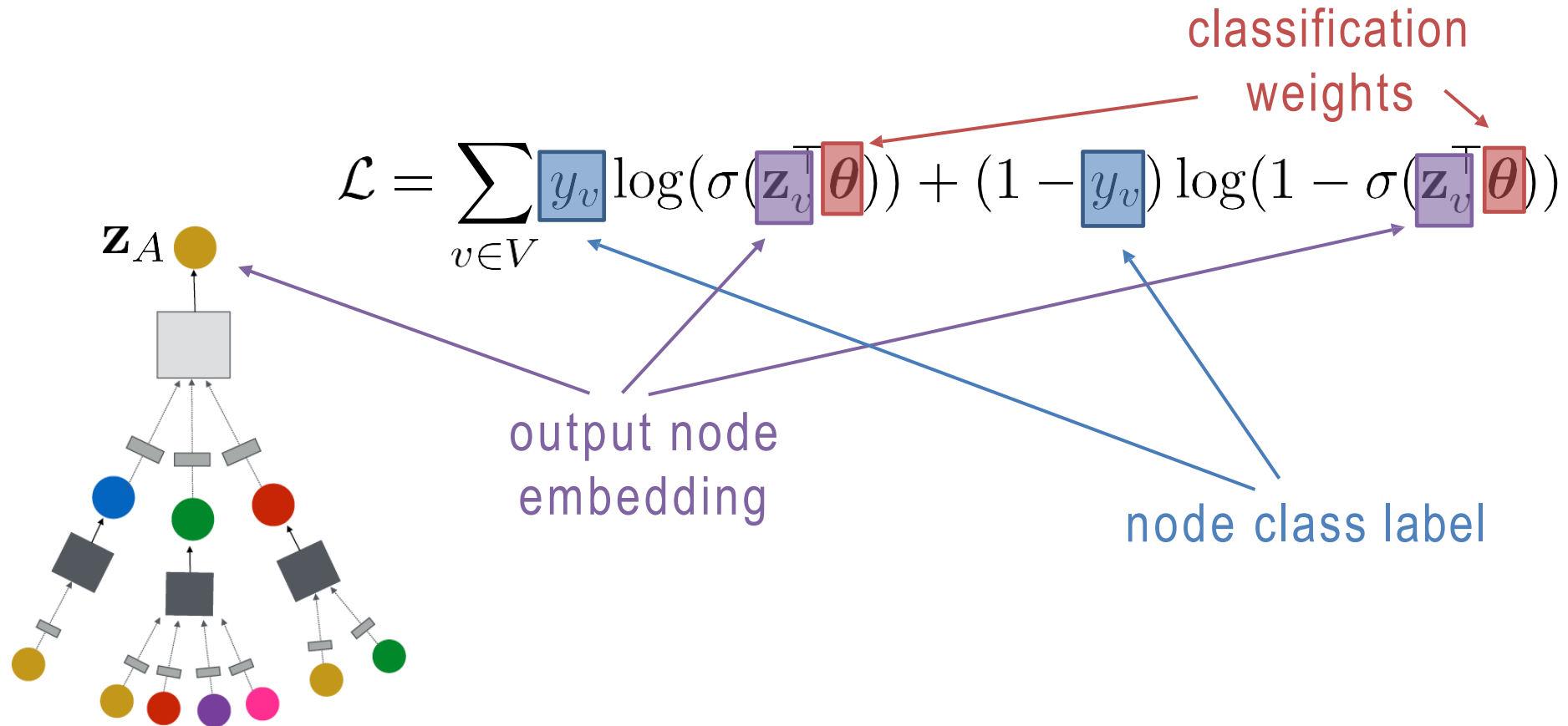
Graph Attention Network (Velickovic et al., 2018)

$$\mathbf{h}_v^k = \sigma \left(\mathbf{W}_k \sum_{u \in N(v) \cup v} \frac{\alpha_{uv} \mathbf{h}_u^{k-1}}{\sqrt{|N(u)| |N(v)|}} \right)$$

Attention weights

Very similar to a self-attention transformer

Graph Neural Nets – Supervised Training

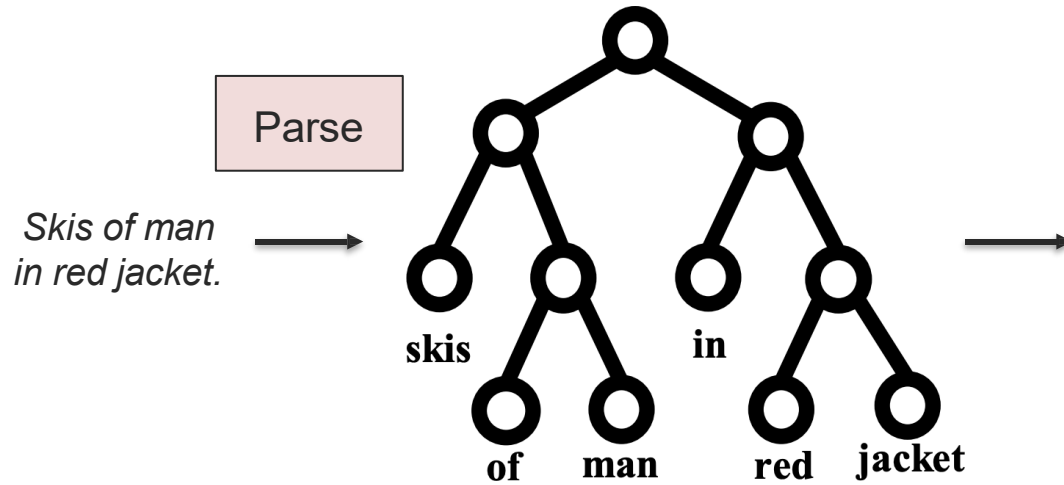


Going Beyond Sequences: Hierarchical Structure

*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

Hierarchical Structure

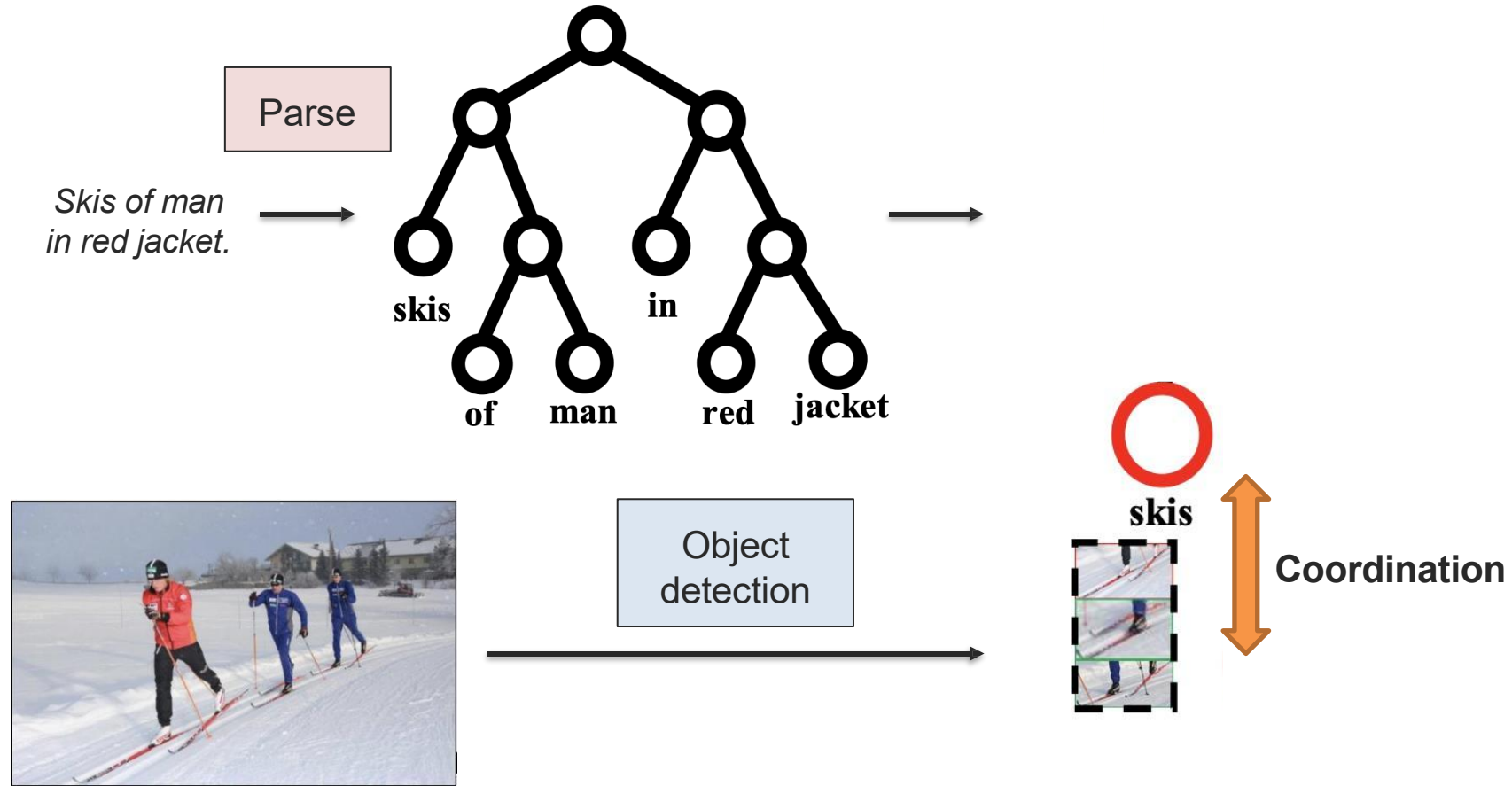
Leverage syntactic structure of language



Object
detection

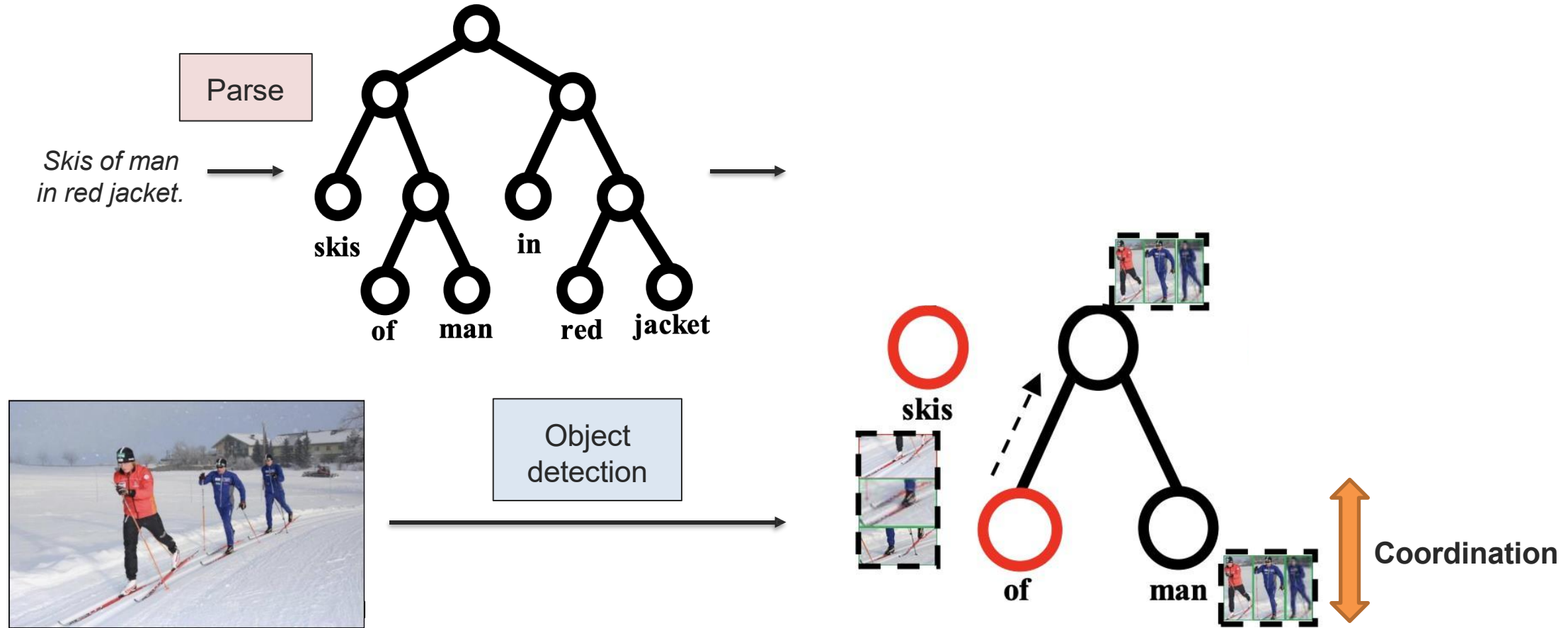
Hierarchical Structure

Leverage syntactic structure of language



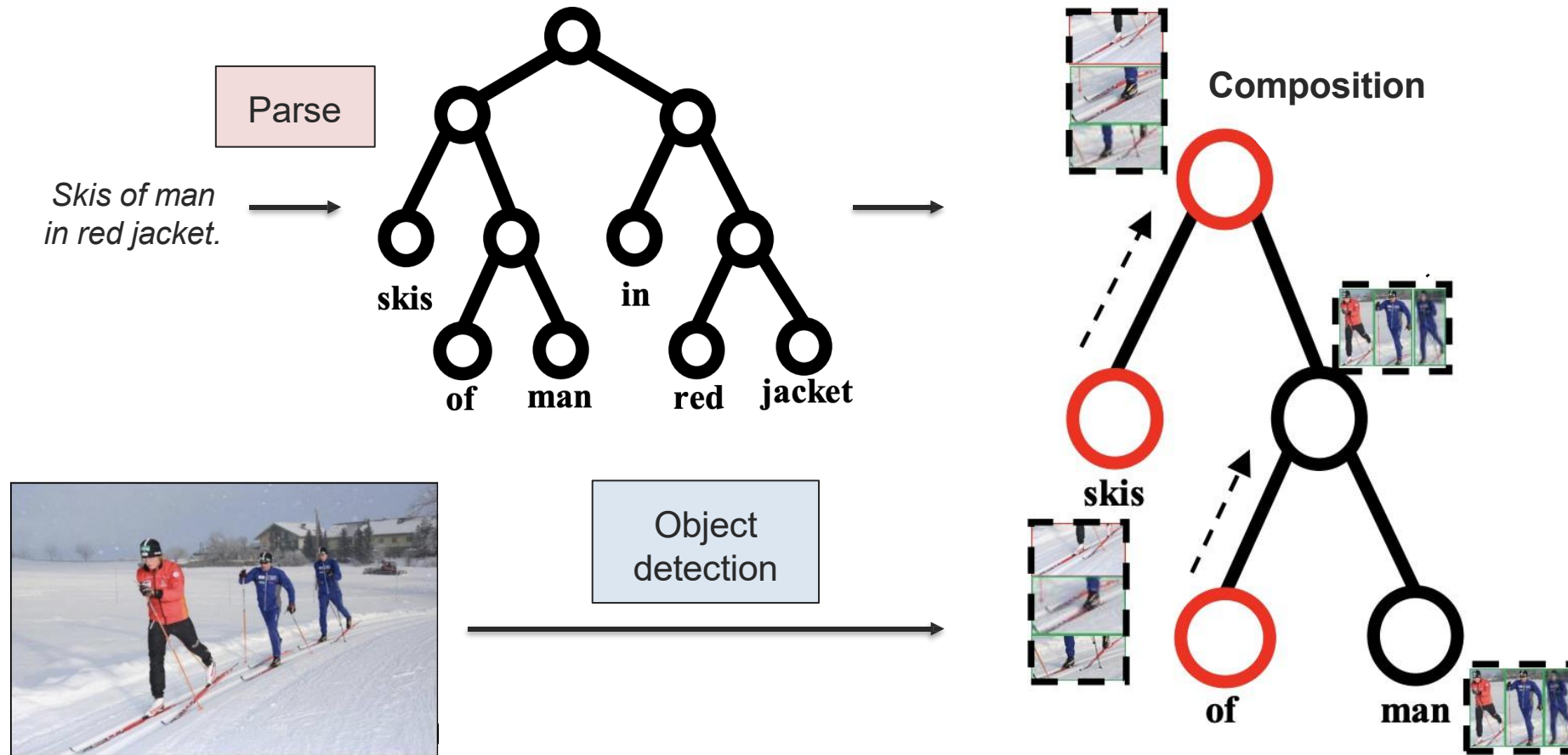
Hierarchical Structure

Leverage syntactic structure of language



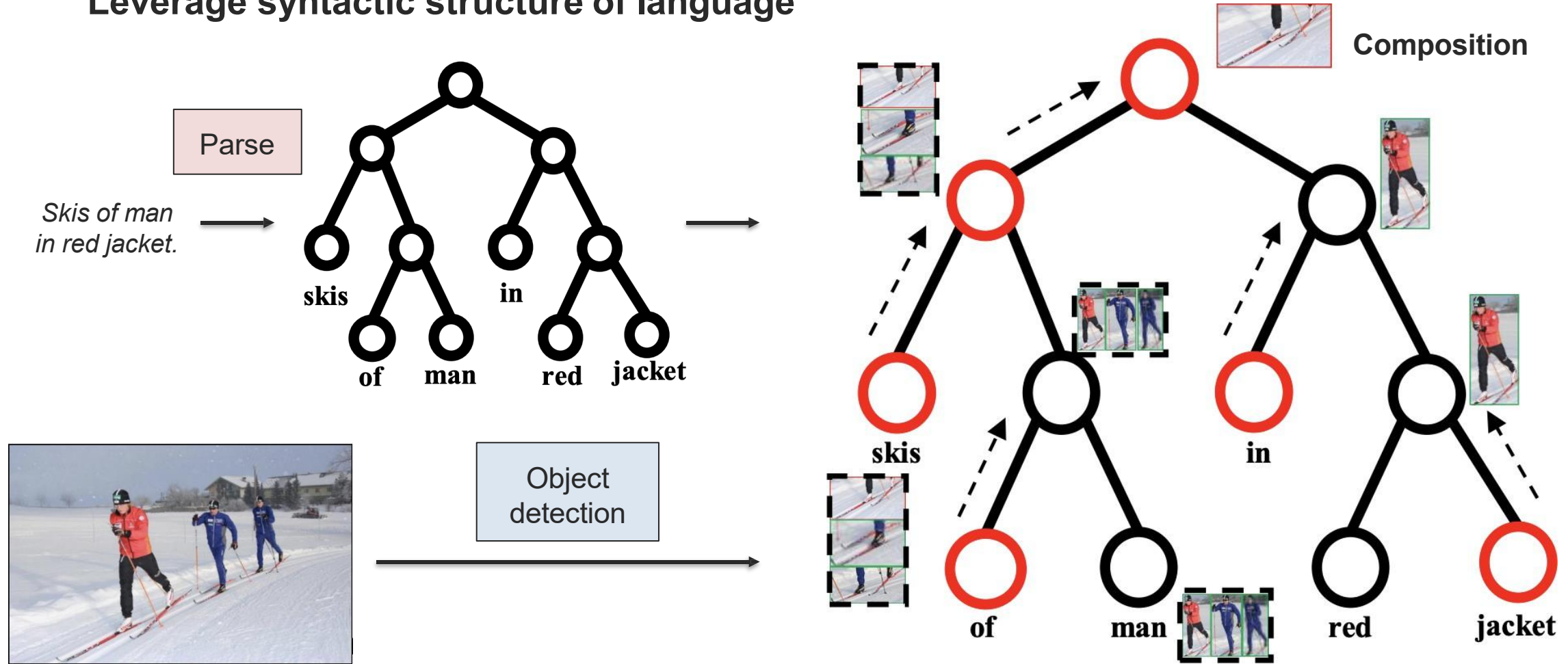
Hierarchical Structure

Leverage syntactic structure of language



Hierarchical Structure

Leverage syntactic structure of language



Hierarchical Structure

Leverage syntactic structure of language

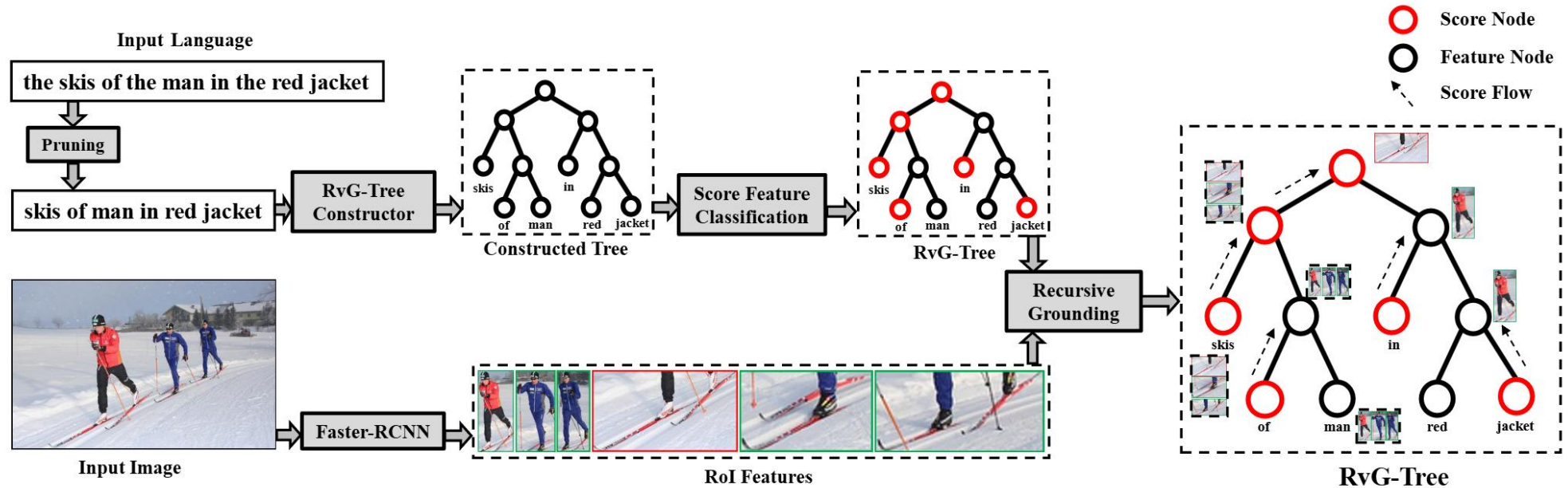
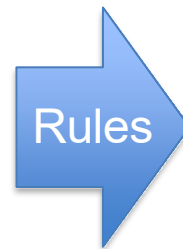
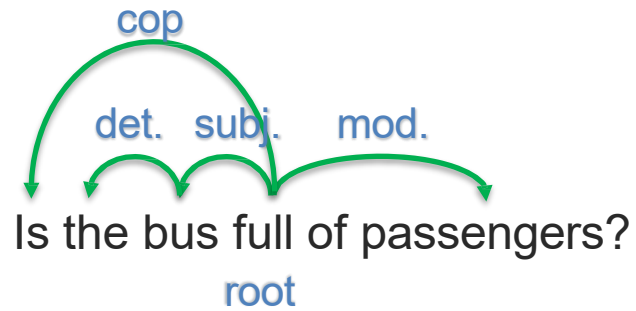


Fig. 2: The overview of using RVG-TREE for natural language grounding. Given a natural language sentence as input, we first prune the sentence and then construct RVG-TREE (Section 3.2). Then, we have a score-feature classifier (Section 3.3) to determine each node as the "score node" or "feature node", where the score node returns the recursive score and the feature node returns the feature (Section 3.3). The final score of the root node is accumulated recursively in a bottom-up fashion (Section 3.3) and the visual region with the highest score is considered as the grounding result. Note that all the nodes can be visualized by the corresponding confidence scores and only qualitative regions are visualized.

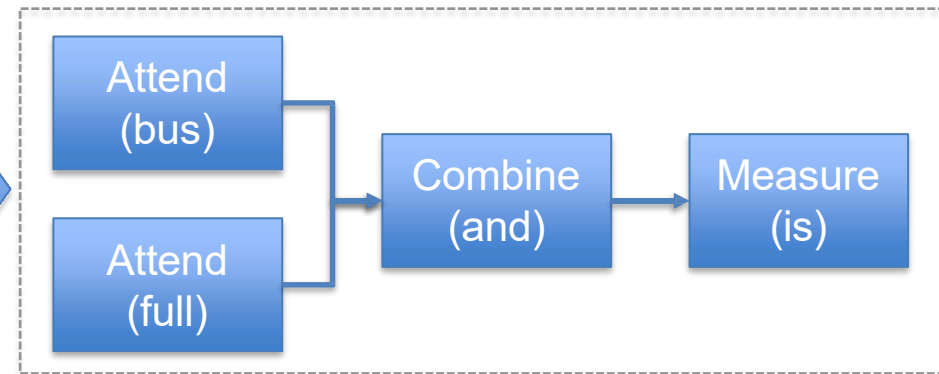
Going Beyond Sequences: Modular Structure

*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

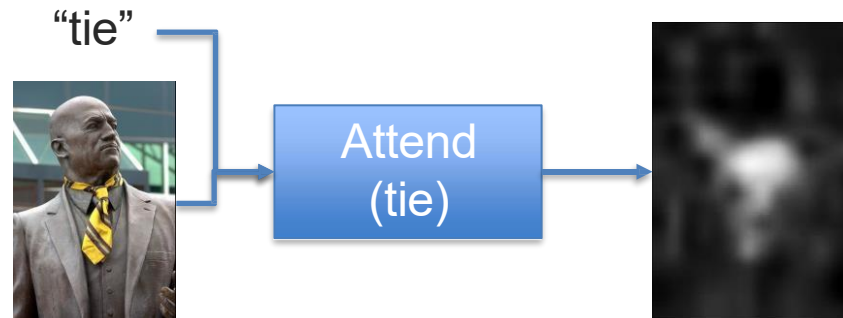
Neural Module Network



Computation layout



Each module work on the attention map(s):

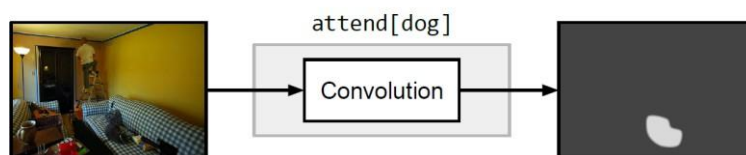


Andreas et al., Deep Compositional Question Answering with Neural Module Networks, 2016

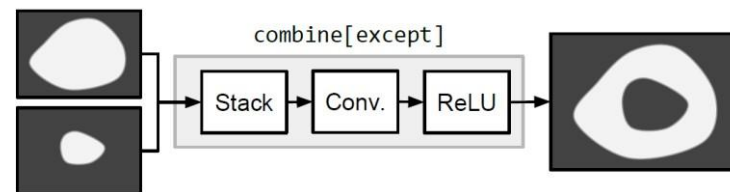
Predefined Set of Modules

1) Analyze the image:

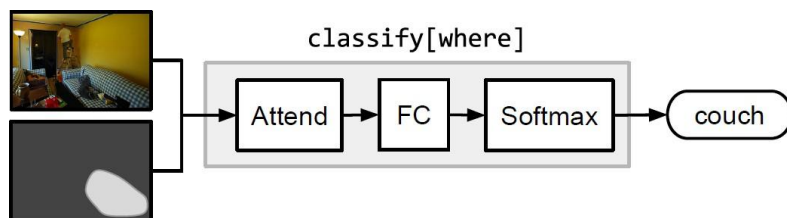
$\text{attend} : \text{Image} \rightarrow \text{Attention}$



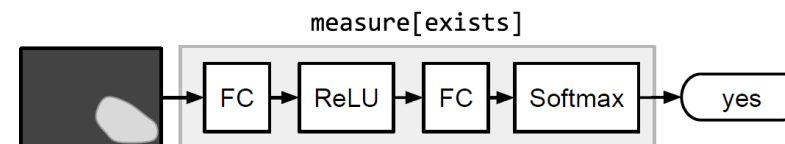
$\text{combine} : \text{Attention} \times \text{Attention} \rightarrow \text{Attention}$



2) Make a prediction

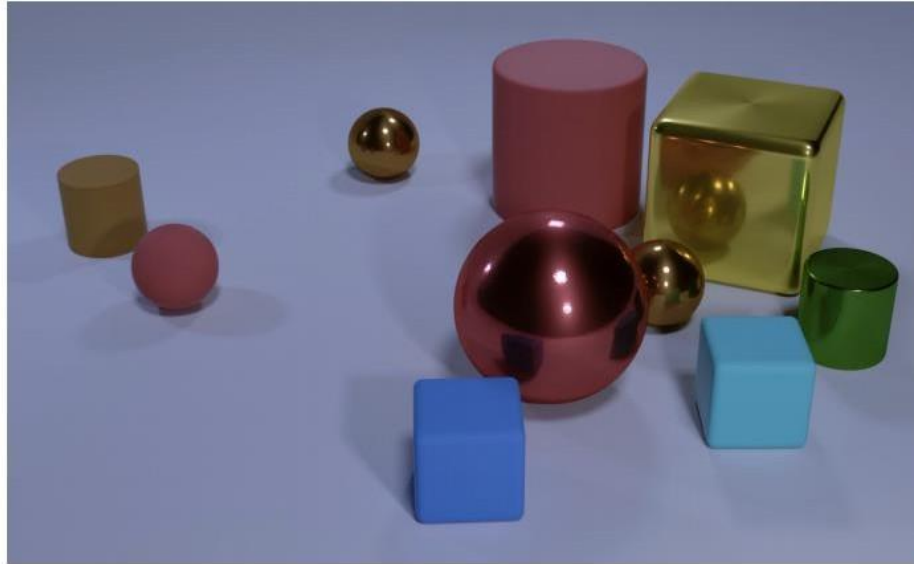


$\text{measure} : \text{Attention} \rightarrow \text{Label}$



CLEVR: Dataset for Visual Reasoning

Perfect for a neural module network!



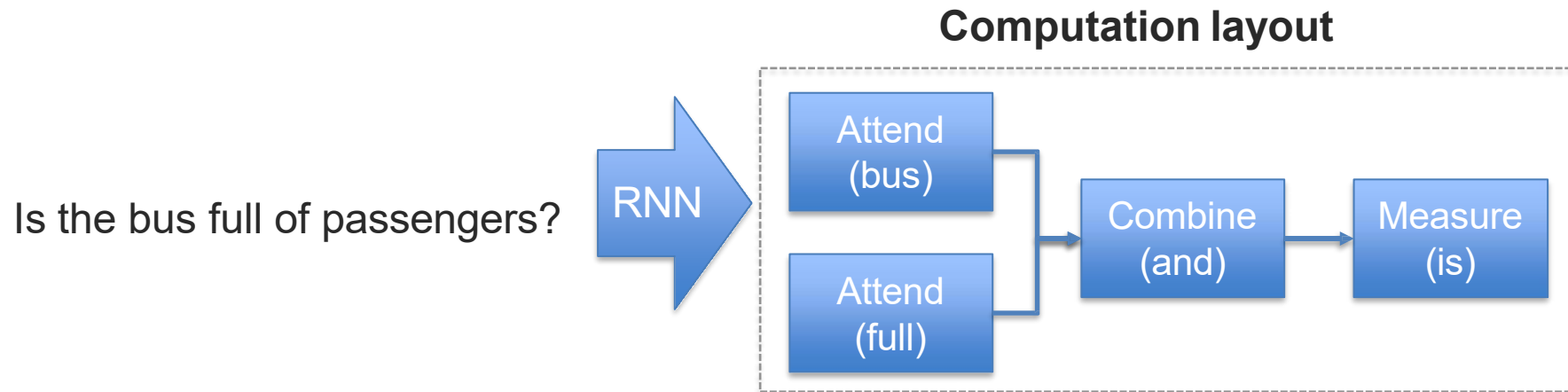
Q: Are there an **equal** number of large things and metal spheres?

Q: What size is the cylinder that is left of the brown metal thing that is left of the big sphere? Q: There is a sphere with the same size as the

metal cube; is it made of the same material as the small red sphere?

Q: How many objects are either small cylinders or metal things?

Module Network V2: End-to-End Learning



No need to parse the question!

No rule-based creation of the layout!

Hu et al., Learning to Reason: End-to-End Module Networks for Visual Question Answering, 2017

Module Network V2: End-to-End Learning

There is a shiny object that is right of the gray metallic cylinder;
does it have the same size as the large rubber sphere?

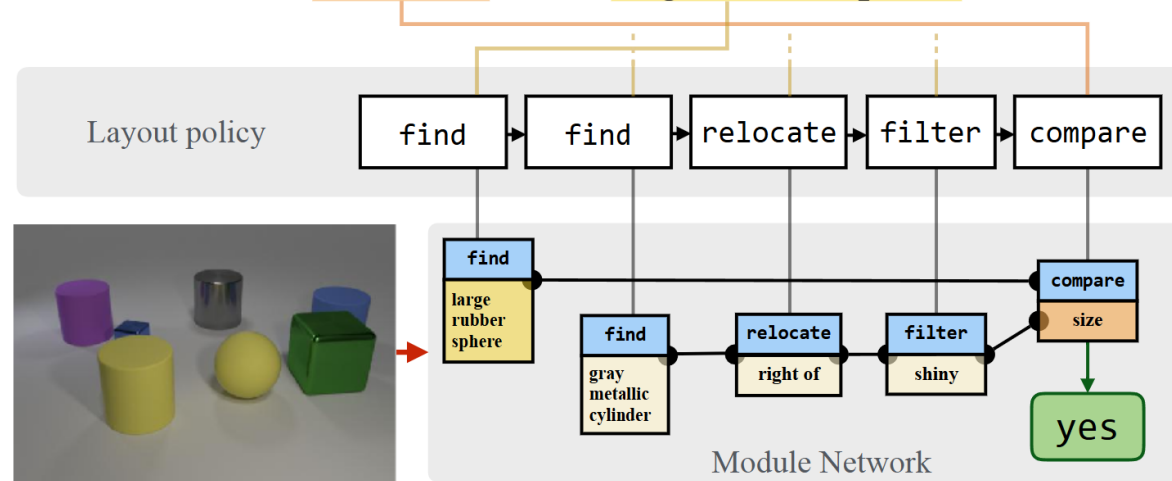
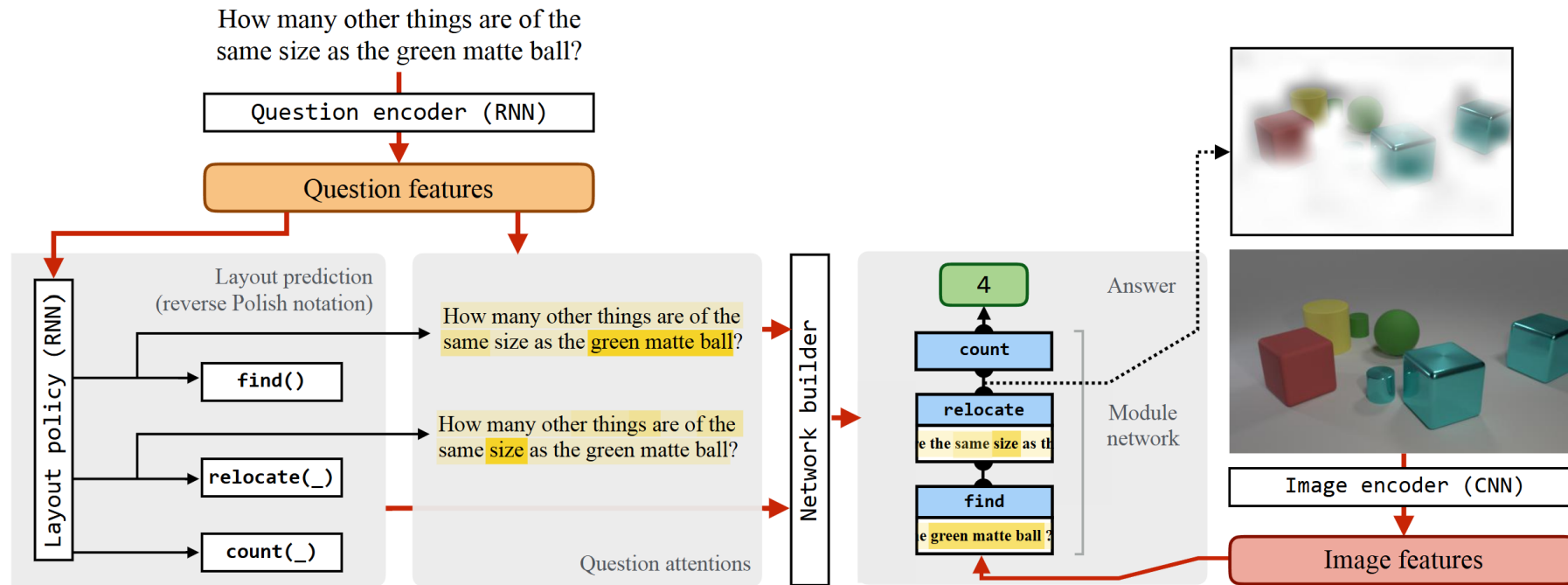


Figure 1: For each instance, our model predicts a computational expression and a sequence of attentive module parameterizations. It uses these to assemble a concrete network architecture, and then executes the assembled neural module network to output an answer for visual question answering. (The example shows a real structure predicted by our model, with text attention maps simplified for clarity.)

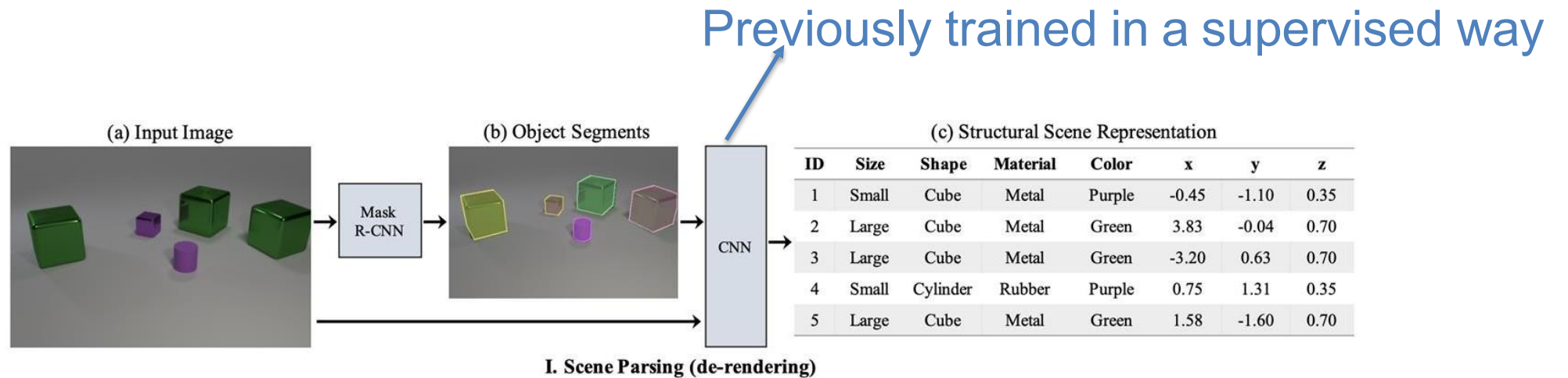
Module Network V2: End-to-End Learning



Model overview. Our approach first computes a deep representation of the question, and uses this as an input to a layout-prediction policy implemented with a recurrent neural network. This policy emits both a sequence of structural actions, specifying a template for a modular neural network in reverse Polish notation, and a sequence of attentive actions, extracting parameters for these neural modules from the input sentence. These two sequences are passed to a network builder, which dynamically instantiates an appropriate neural network and applies it to the input image to obtain an answer.

Module Network V3: Neural-symbolic VQA

1) Image Attributes

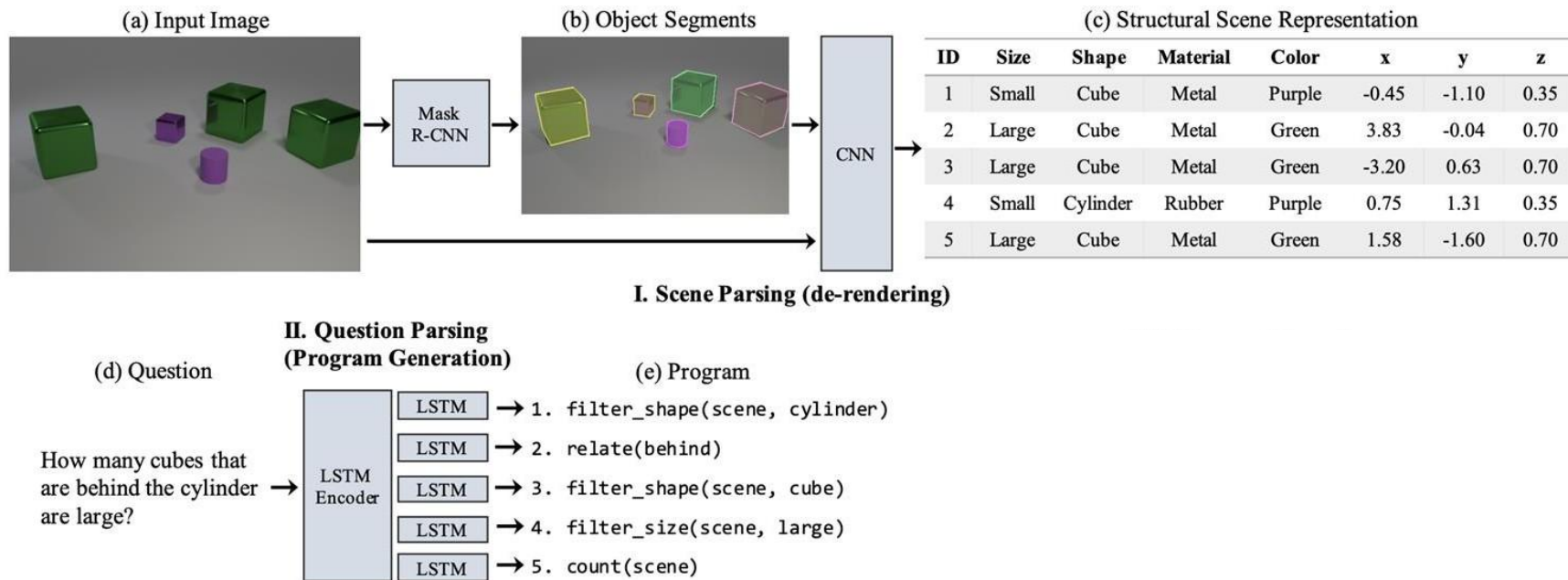


Kexin Yi, et al. "Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding." Neurips 2018

Module Network V3: Neural-symbolic VQA

2) Parsing questions into programs

Similar to neural module networks

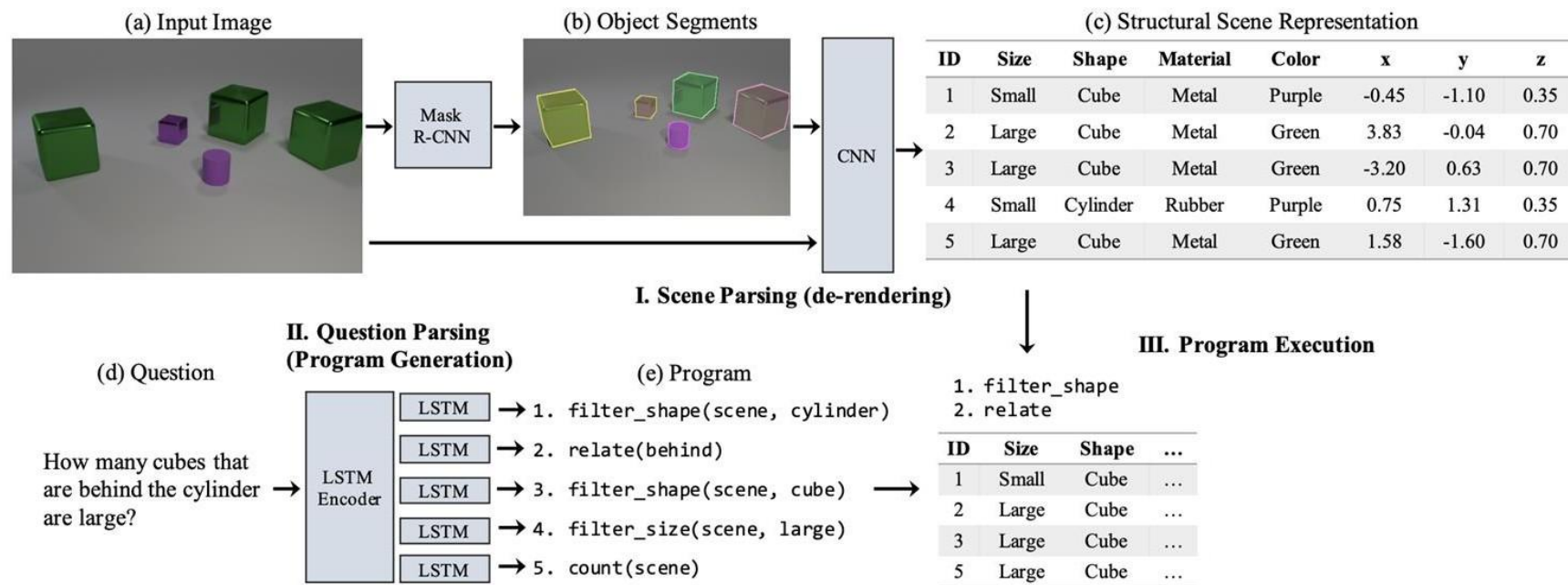


Kexin Yi, et al. "Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding." Neurips 2018

Module Network V3: Neural-symbolic VQA

3) Program execution

Execution of the program is somewhat easier given the “symbolic” representation of the image



Kexin Yi, et al. “Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding.” Neurips 2018

Module Networks V4: The Neural State Machine

How to solve this question
using visual reasoning?



What is the *red* fruit inside the *bowl*
to the *right* of the *coffee maker*?

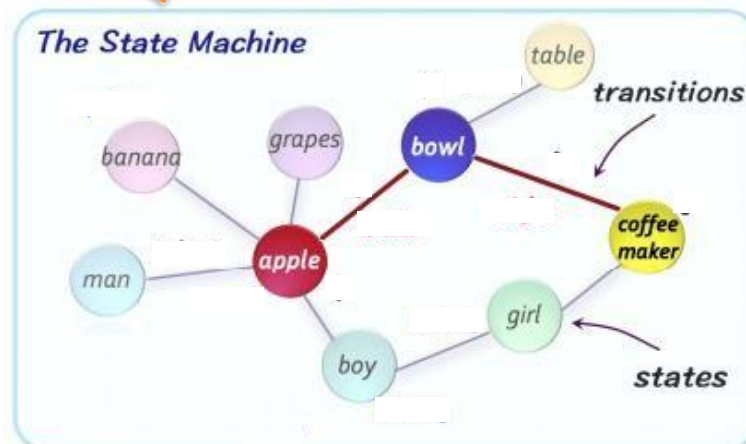
1. Given an **image**, generate a probabilistic **scene graph** that captures the semantic concepts.
2. Treat the graph as a **state machine** and simulate iterative computation over it to *answer questions or draw inferences*.
3. Natural language questions are translated into *soft instructions* and used to perform sequential reasoning over the scene graph/state machine.

Module Networks V4: The Neural State Machine

Detect objects and create proximity graph



What is the *red fruit* inside the *bowl*
to the *right* of the *coffee maker*?



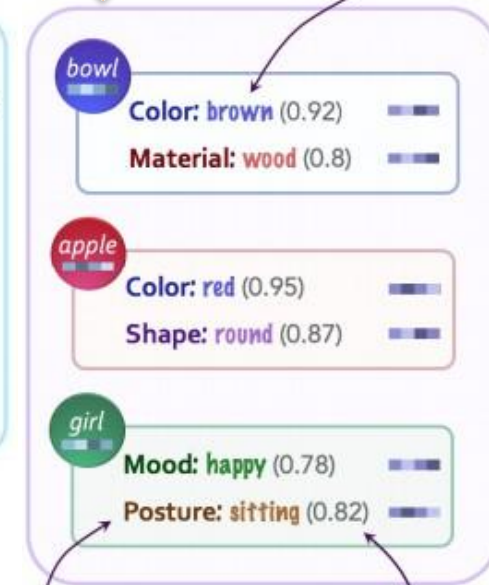
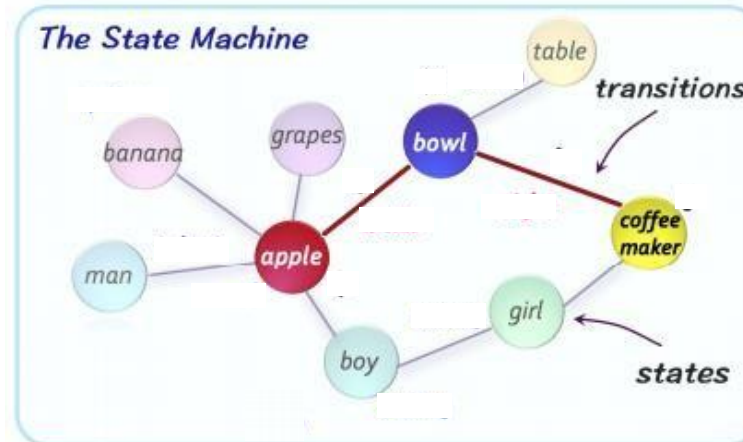
Module Networks V4: The Neural State Machine

Pre-trained an alphabet of concepts
(Visual Genome)

↓ *alphabet (concepts)*



What is the *red* fruit inside the *bowl*
to the *right* of the *coffee maker*?

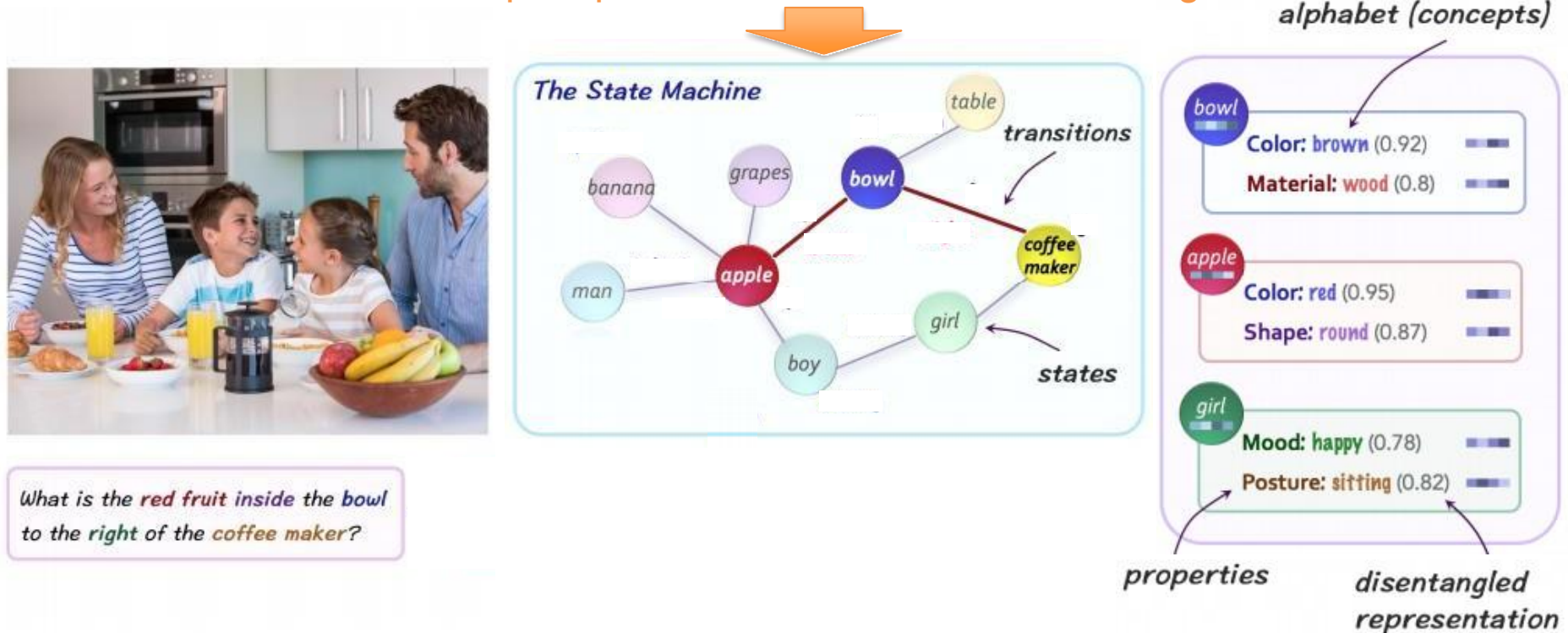


Manually grouped
by "properties"

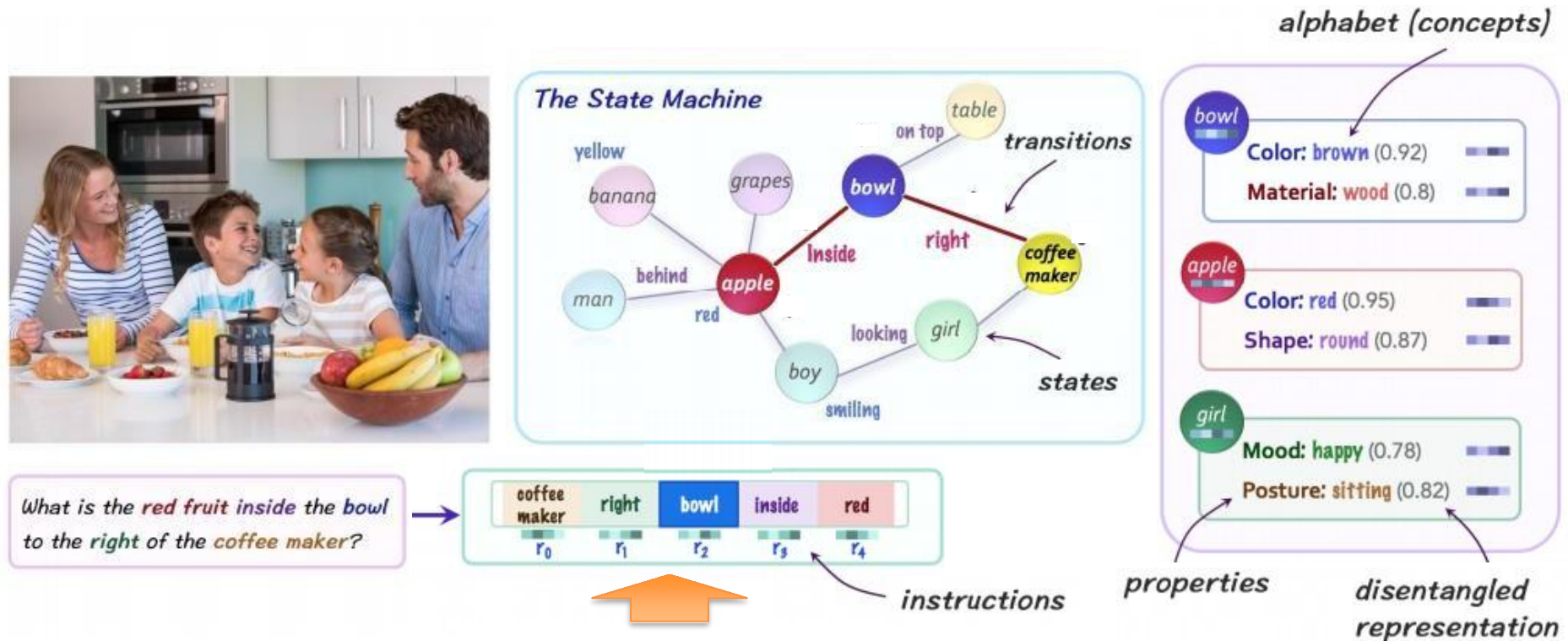
Probabilities
computed at
runtime for each
object instance

Module Networks V4: The Neural State Machine

Predefined an alphabet of relations
and compute probabilities for each directed edges



Module Networks V4: The Neural State Machine

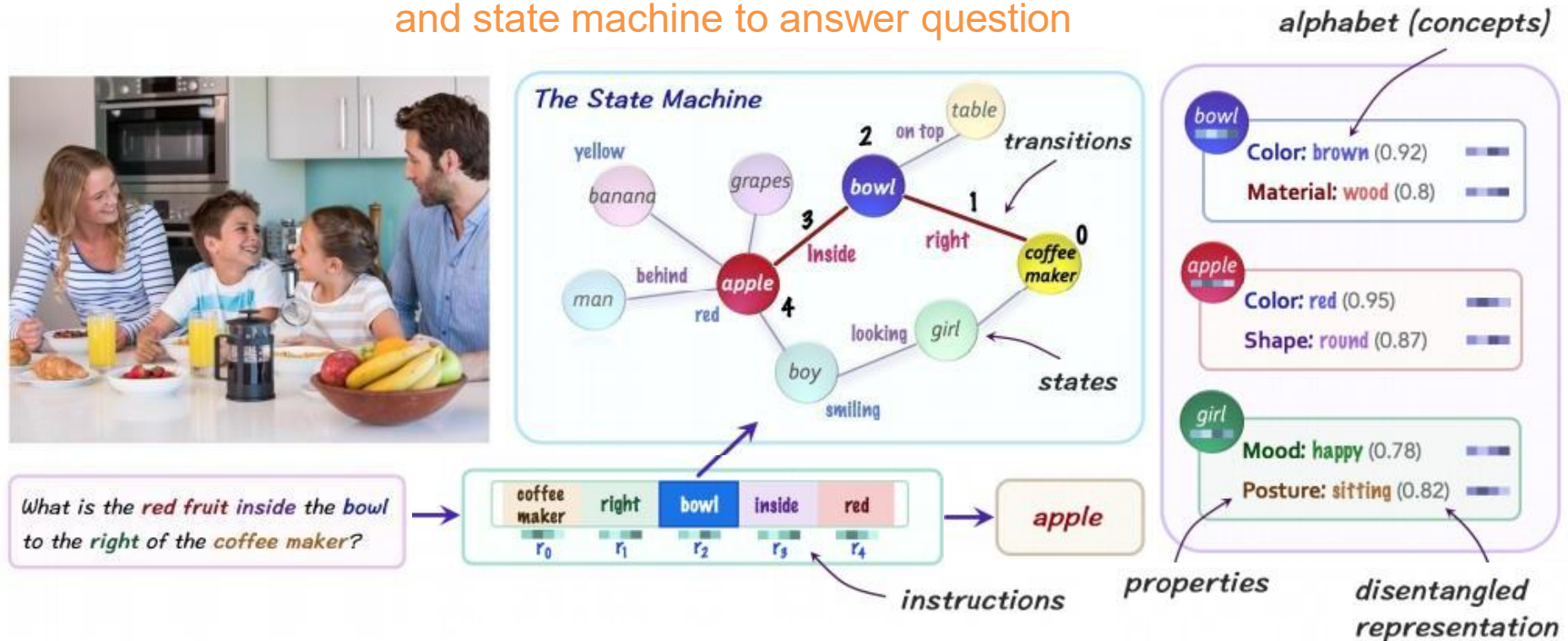


Translate each word in a concept-based representation
and group in a fixed number of instruction steps

Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine." NeurIPS 2019

Module Networks V4: The Neural State Machine

Finally, perform reasoning using instructions and state machine to answer question



References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University